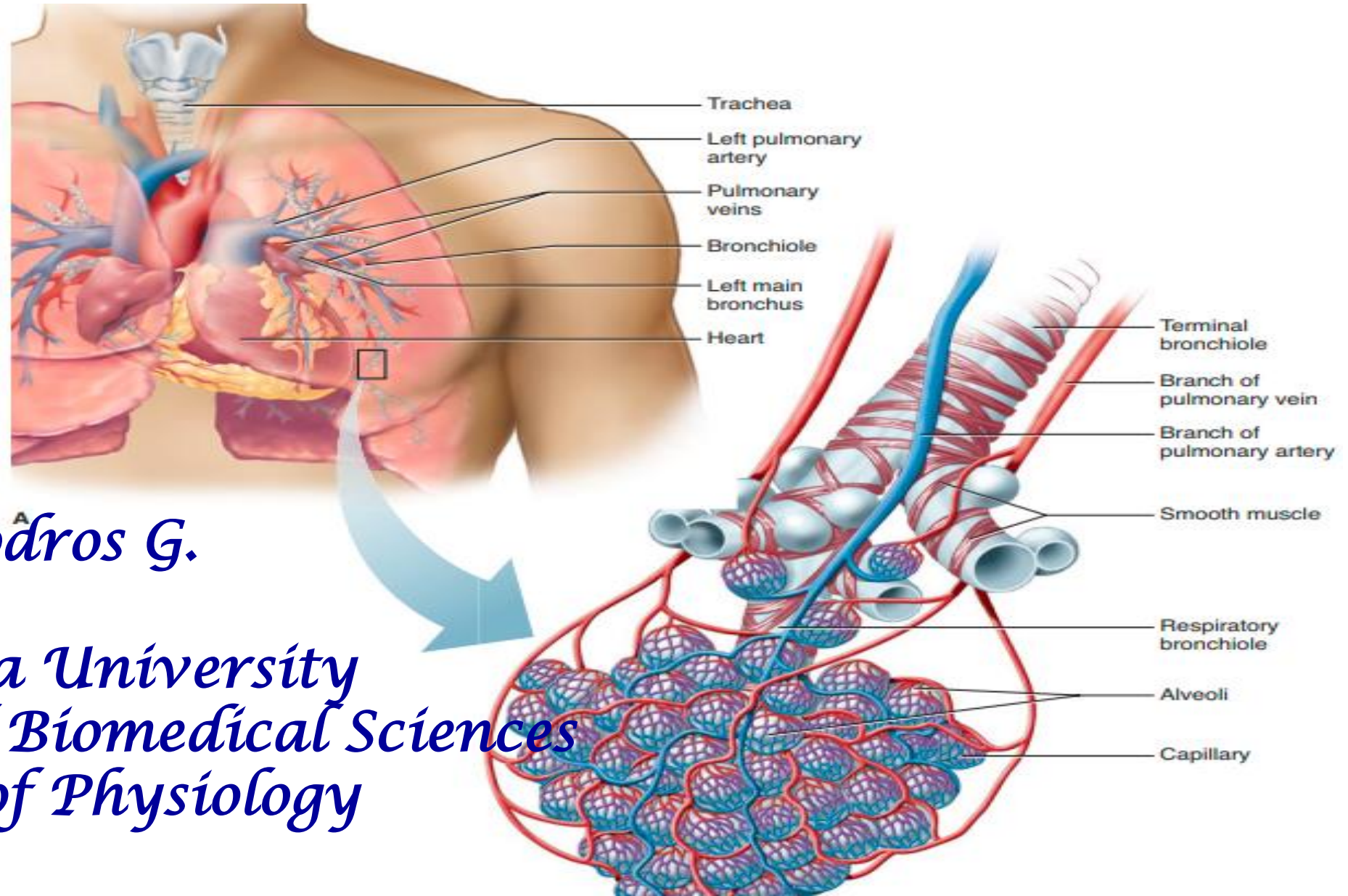


Physiology of Respiratory System



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Principles of Gas Exchange

- ☐ Diffusion of Oxygen
- ☐ Diffusion of Carbon Dioxide

Diffusion of Gases

- ❑ Oxygen and CO₂ move passively from a region of higher to one of lower concentration.
- ❑ The **rate of gas diffusion** (VG) through the tissue is **directly proportional to:**
 - ⊗ Tissue surface area
 - ⊗ Partial pressure difference (P₁-P₂) b/n the two sides
 - ⊗ Fluid temperature
 - ⊗ Solubility coefficient of the gas
- ❑ **Inversely proportional to:**
 - ⊗ Fluid viscosity
 - ⊗ Thickness of the tissue
 - ⊗ Square root of molecular weight of the gas

Fick's Law of Diffusion

$$VG \propto \frac{\Delta PTA}{\eta L \sqrt{MW}}$$

Diffusion of Gases

- Gases diffuse from high pressure → low pressure
 - ☒ Between lungs and pulmonary capillary
 - ☒ Between systemic capillary and tissue cells

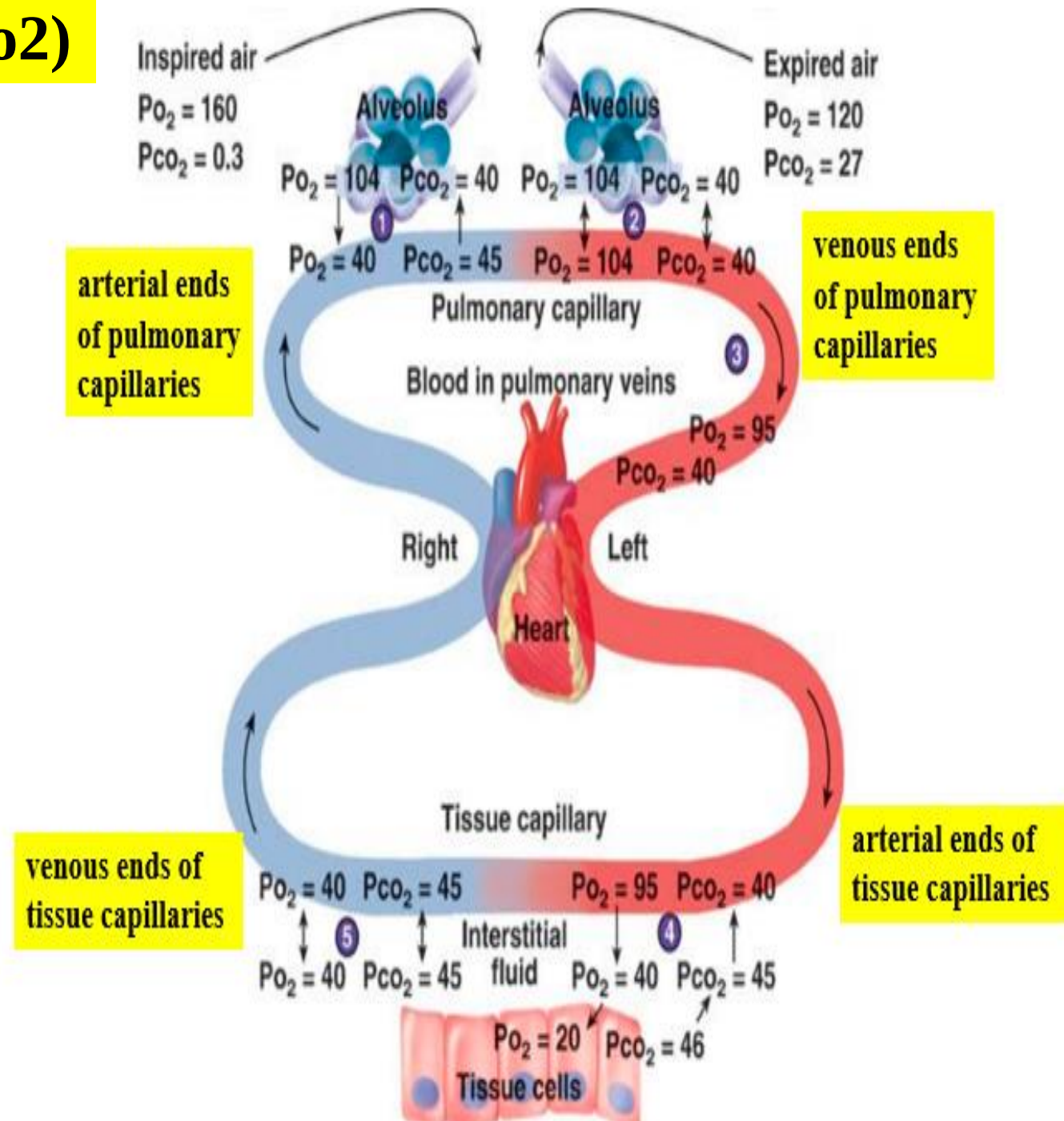
- Oxygen and CO₂ exchange across pulmonary and systemic capillaries caused by **partial pressure gradients**
 - **Oxygen diffuses** from the alveoli into the pulmonary capillaries because the **Po₂ in the alveoli is greater than in the pulmonary capillaries**
 - In contrast, **Carbon Dioxide diffuses** from the pulmonary capillaries into the alveoli because the **Pco₂ is greater in the pulmonary capillaries than in the alveoli.**

Partial Pressure of Oxygen (P_{O_2})

- ❑ The P_{O_2} within the **alveoli** approximately **104 mm Hg**, whereas the P_{O_2} in blood flowing into the **pulmonary capillaries** is approximately **40 mm Hg**.

⇒ Consequently, **oxygen** diffuses down its partial pressure gradient from the **alveoli** into the **pulmonary capillary**.

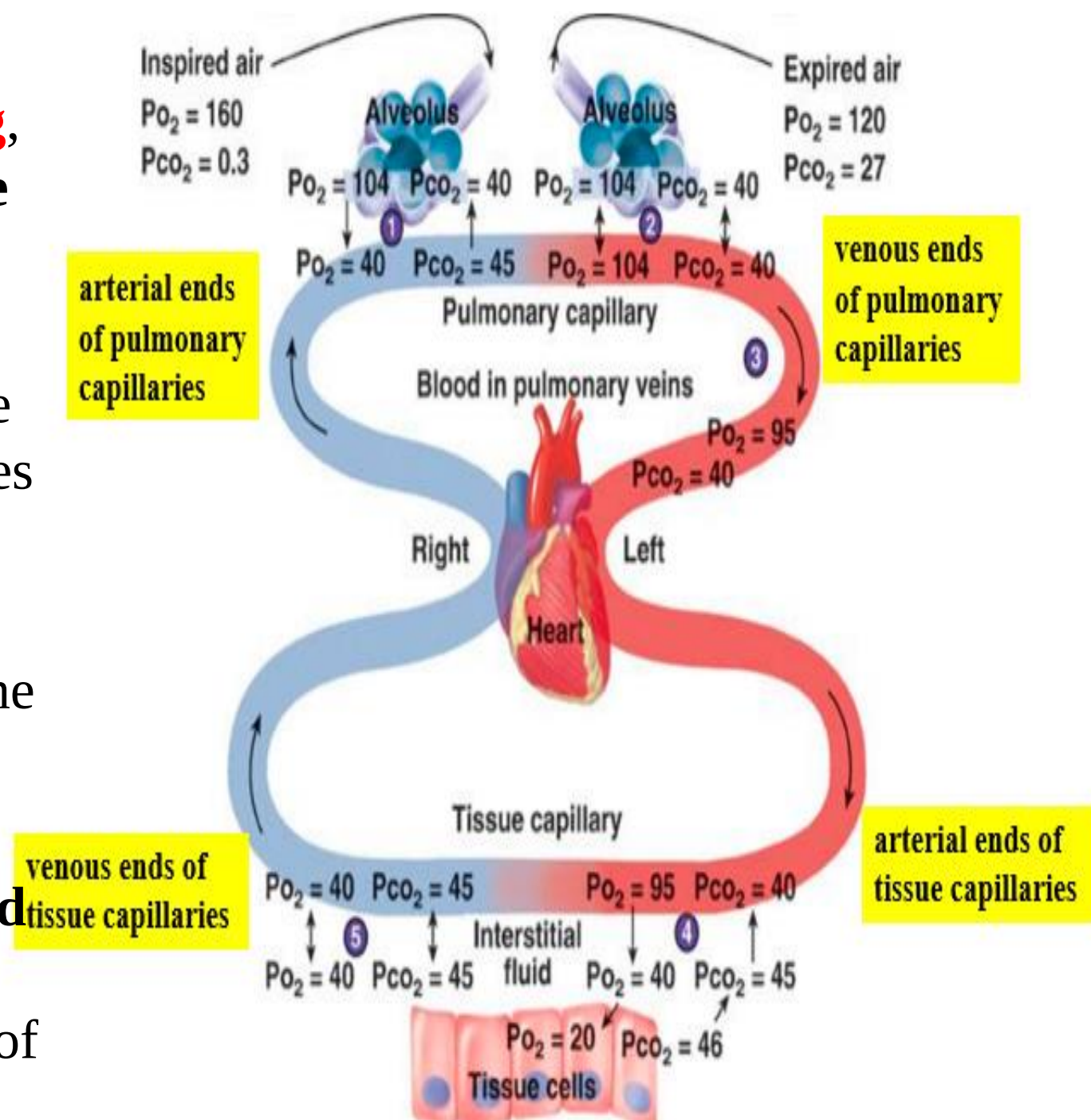
- ❑ By the time blood flows through the pulmonary capillary beds, an **equilibrium** has been achieved, and the P_{O_2} in the blood is **104 mm Hg**, which is equivalent to the P_{O_2} in the **alveoli**



- ❑ Blood leaving the **Pulmonary Capillaries** has a **Po₂ of 104 mm Hg**, **BUT** blood leaving the **lungs**; in the **Pulmonary Veins** has a **Po₂ of 95 mm Hg**

⇓ This **decrease in Po₂** occurs b/c the blood from the pulmonary capillaries mixes with deoxygenated blood called **shunted blood**.

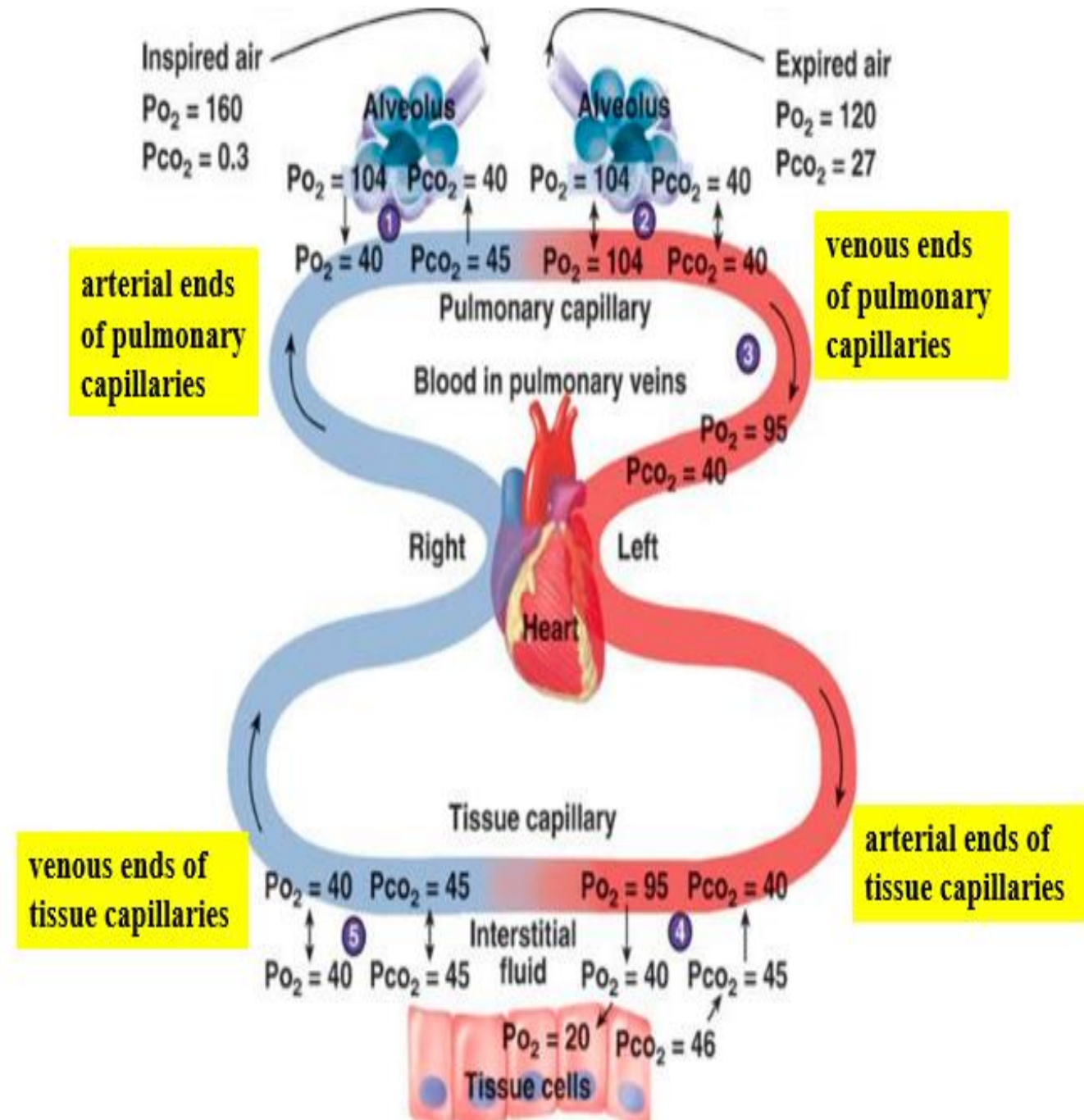
- ☒ One source of shunted blood is the **deoxygenated blood from bronchial circulation**.
- ☒ The other source is **deoxygenated blood from pulmonary circulation**. Normally, **1%–2%** of cardiac output is shunted blood



❑ Blood that enters the **arterial end of tissue capillaries** has a P_{O_2} of approximately **95 mm Hg**. The P_{O_2} of the interstitial fluid, in contrast, is close to **40 mm Hg** and **20 mm Hg** in the individual cells.

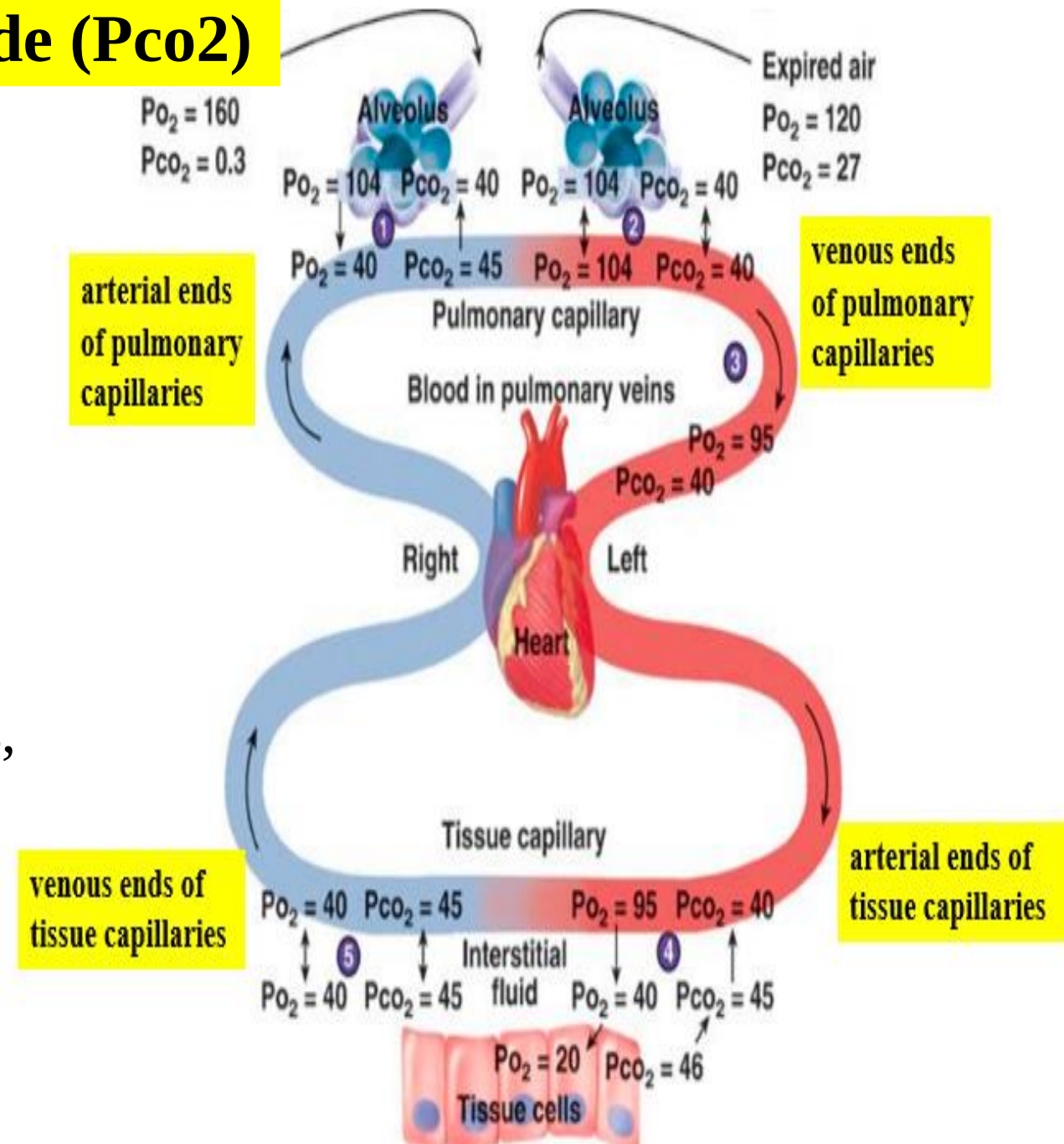
⇒ **Oxygen diffuses** from the **tissue capillaries to interstitial fluid** and **from the interstitial fluid into the cells**, where it is used in aerobic metabolism.

☒ Because the cells use oxygen continuously, a **constant partial pressure** exists for **oxygen from the tissue capillaries to cells**

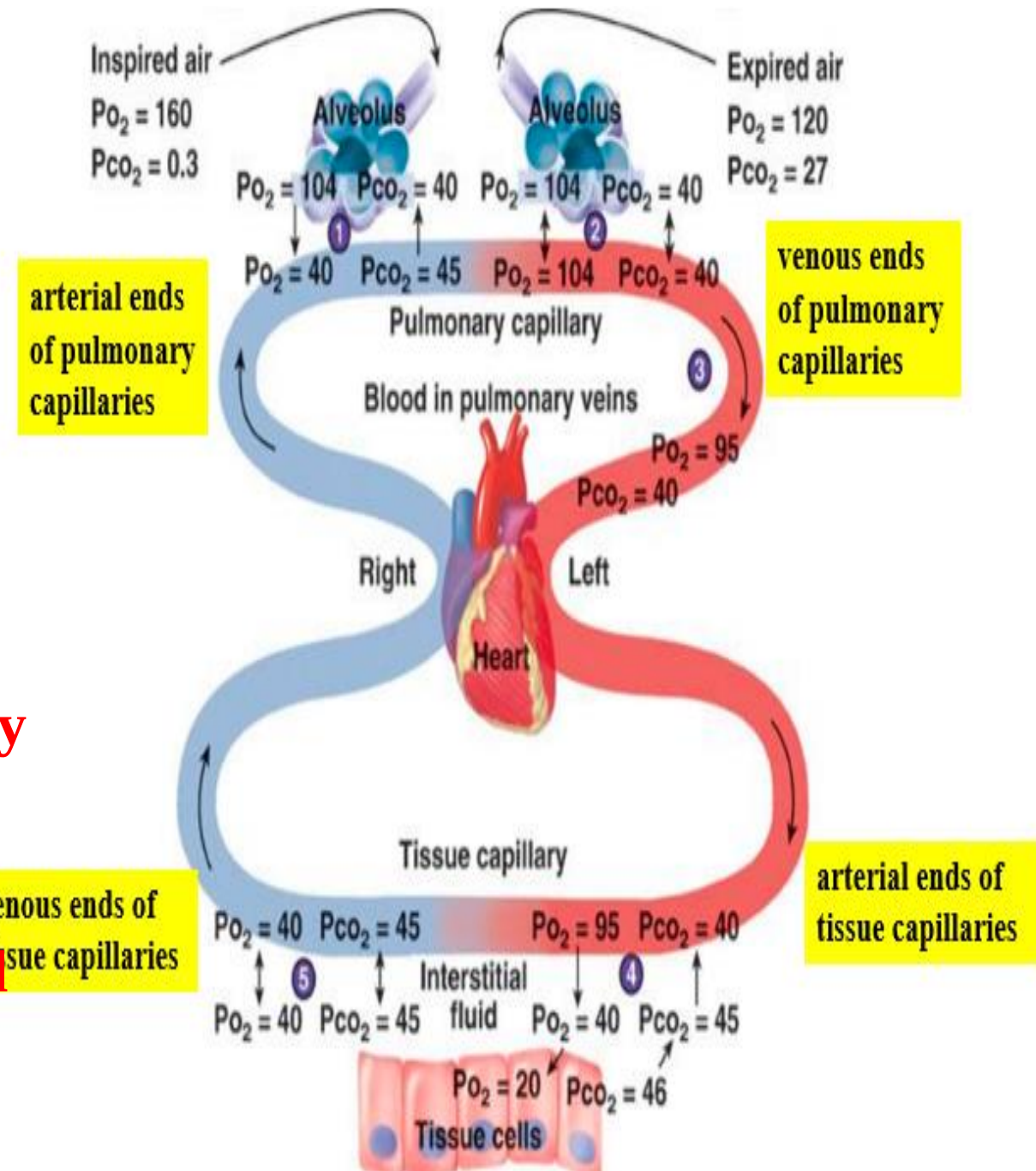


Partial Pressure of Carbon Dioxide (P_{CO_2})

- ❑ **CO₂** is produced as a by-product of cellular respiration, and a **partial pressure** gradient is **established** from tissue cells to tissue capillaries.
- ❑ The **intracellular P_{CO_2}** is approximately **46 mm Hg**, and the **interstitial fluid P_{CO_2}** is approximately **45 mm Hg**.
- ❑ At the **arterial end of tissue capillaries**, the **P_{CO_2} is close to 40 mm Hg**.
 - ⇒ As blood flows through the **tissue capillaries**, **CO₂ diffuses** from a higher P_{CO_2} to a lower P_{CO_2} until an **equilibrium in P_{CO_2} is established**.



- ❑ At the **venous end of the capillaries**, blood has a **P_{CO_2} of 45 mm Hg**.
- ⇒ After blood leaves the venous end of the capillaries, it is transported to the lungs.
- ❑ At the **arterial end of the pulmonary capillaries**, the **P_{CO_2} is 45 mm Hg**.
- ❑ Because the **P_{CO_2} is 40 mm Hg** in the **alveoli**, **CO_2 diffuses** from the **pulmonary capillaries** into the **alveoli**.
- ❑ At the **venous end of the pulmonary capillaries**, the **P_{CO_2} has again decreased to 40 mm Hg**.



1. **Oxygen** diffuses into the **arterial ends of pulmonary capillaries**, and **CO₂** diffuses into the **alveoli** b/c of differences in partial pressures

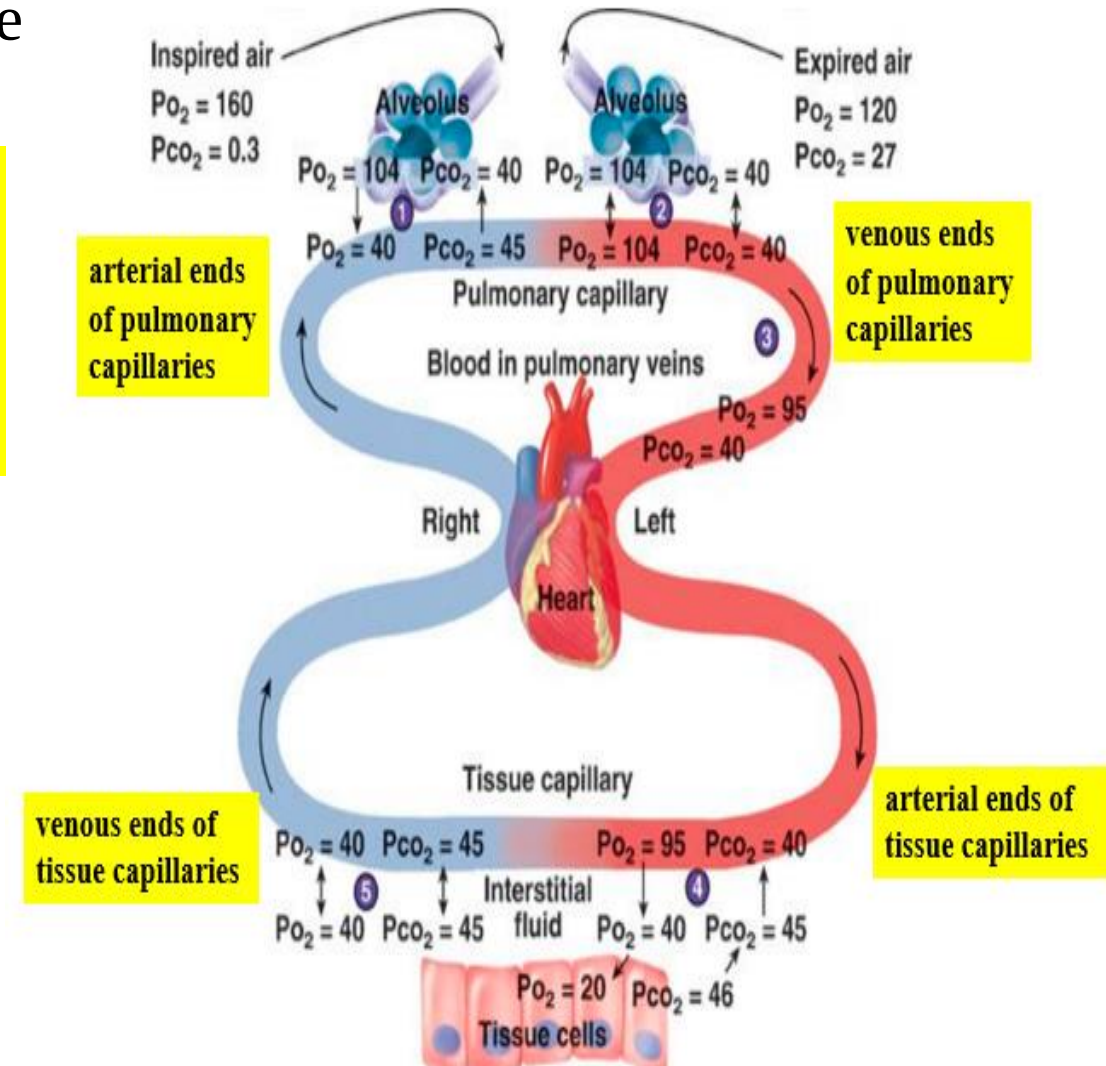
2. As a result of diffusion; at the **venous ends of pulmonary capillaries**, the **PO₂** in the blood is equal to the **PO₂** in the alveoli, and the **PCO₂** in the blood is equal to the **PCO₂** in the alveoli.

3. The **PO₂** in the pulmonary veins is less than in the pulmonary capillaries b/c of mixing with deoxygenated blood from veins draining the bronchi and bronchioles.

4. Oxygen diffuses out of the **arterial ends of tissue capillaries**, and **CO₂** diffuses out of the tissue b/c of differences in partial pressures

5. As a result of diffusion; at the **venous ends of tissue capillaries**, the **PO₂** in the blood is equal to the **PO₂** in the tissue, and the **PCO₂** in the blood is equal to the **PCO₂** in the tissue.

Summary of Partial Pressure of Oxygen and Carbon Dioxide on diffuses



Principles of Gas Transport

- ☐ **Oxygen Transport**

- ☐ **Carbon Dioxide Transport**

Oxygen Transport

❑ Most of the oxygen transported in the blood from the **lungs** to the **tissues** is by **hemoglobin in red blood cells**

❑ **Oxygen is transported :**

1. Hemoglobin (98.5%) and
2. Dissolved in the plasma (1.5%)

❑ Combination of oxygen with hemoglobin is reversible.

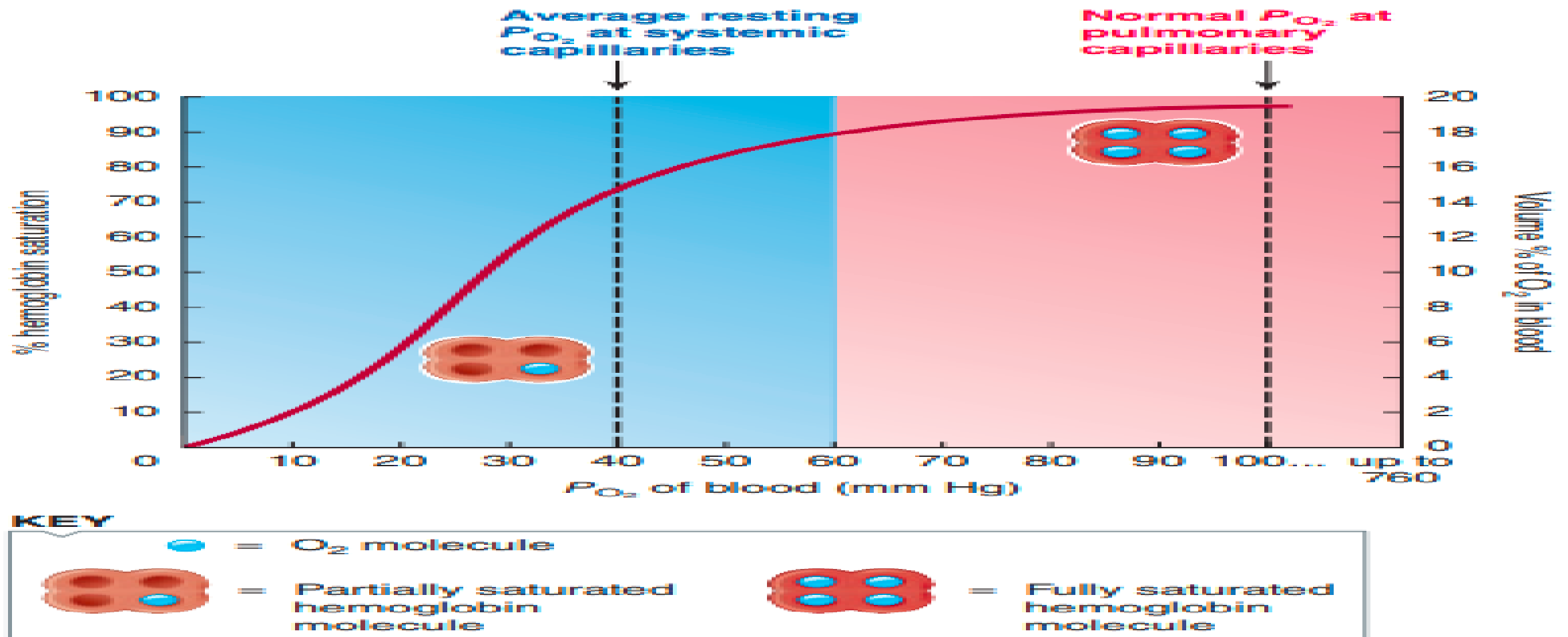
⇒ In the **Pulmonary Capillaries**, oxygen binds to hemoglobin; in the **Tissue Spaces**, oxygen diffuses away from hemoglobin and enters the tissues.

■ The hemoglobin is 100% saturated with oxygen, **when the four oxygen molecules are bound to each hemoglobin molecule.**

⇒ There are four heme groups in a hemoglobin molecule, and oxygen molecule is bound to each heme group.

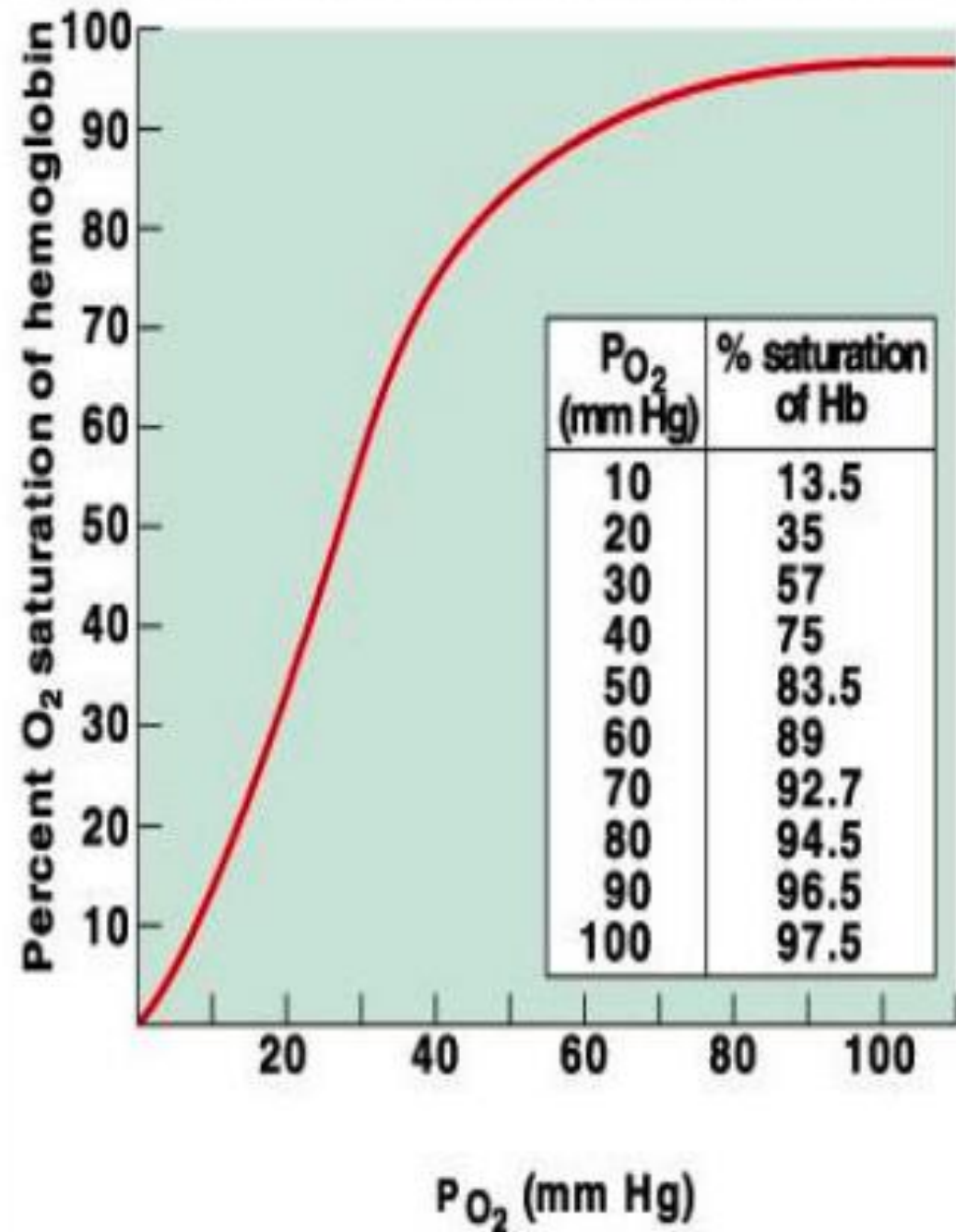
Hemoglobin is partially saturated, when one to three of the hemes are bound to oxygen.

⇒ e.g., Hemoglobin is 50% saturated with oxygen when there is two oxygen molecules bound to each hemoglobin molecule



Oxyhemoglobin Dissociation Curve

- ❑ **Oxygen-hemoglobin dissociation curve** shows that hemoglobin is completely saturated when **P_{O2} is 80 mmHg or above**.
- ❑ At lower partial pressures, the hemoglobin increase the releases of oxygen
- ❑ In addition to **P_{O2}, blood pH, CO₂, and temperature** influence
 - The oxygen binds to hemoglobin



■ **Shift the Curve to the Left** because of

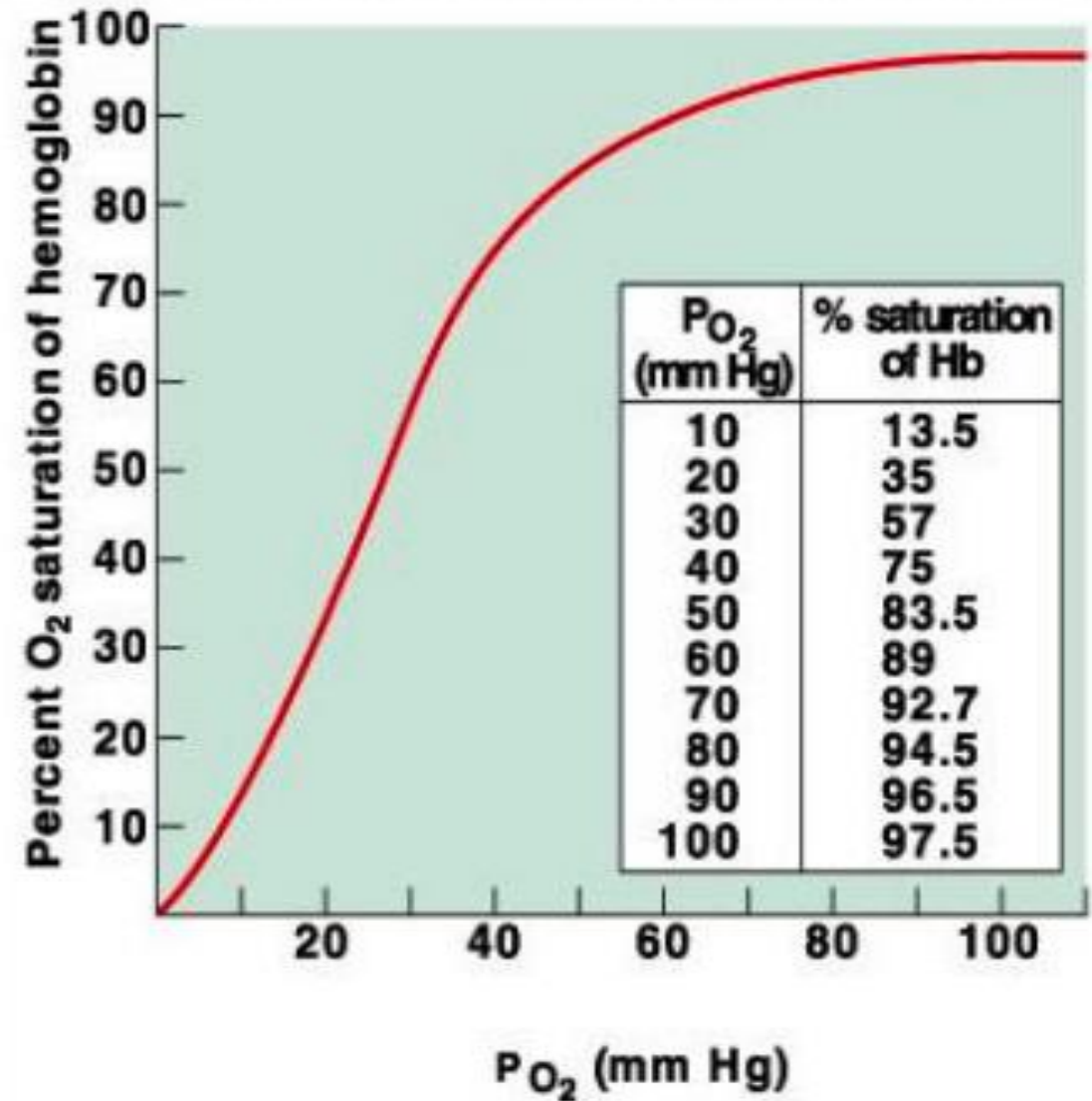
- ☐ An increase p^H
- ☐ A decrease CO_2
- ☐ A decrease in temperature

☒ Results in; **increased** the ability of hemoglobin to **binds with oxygen**

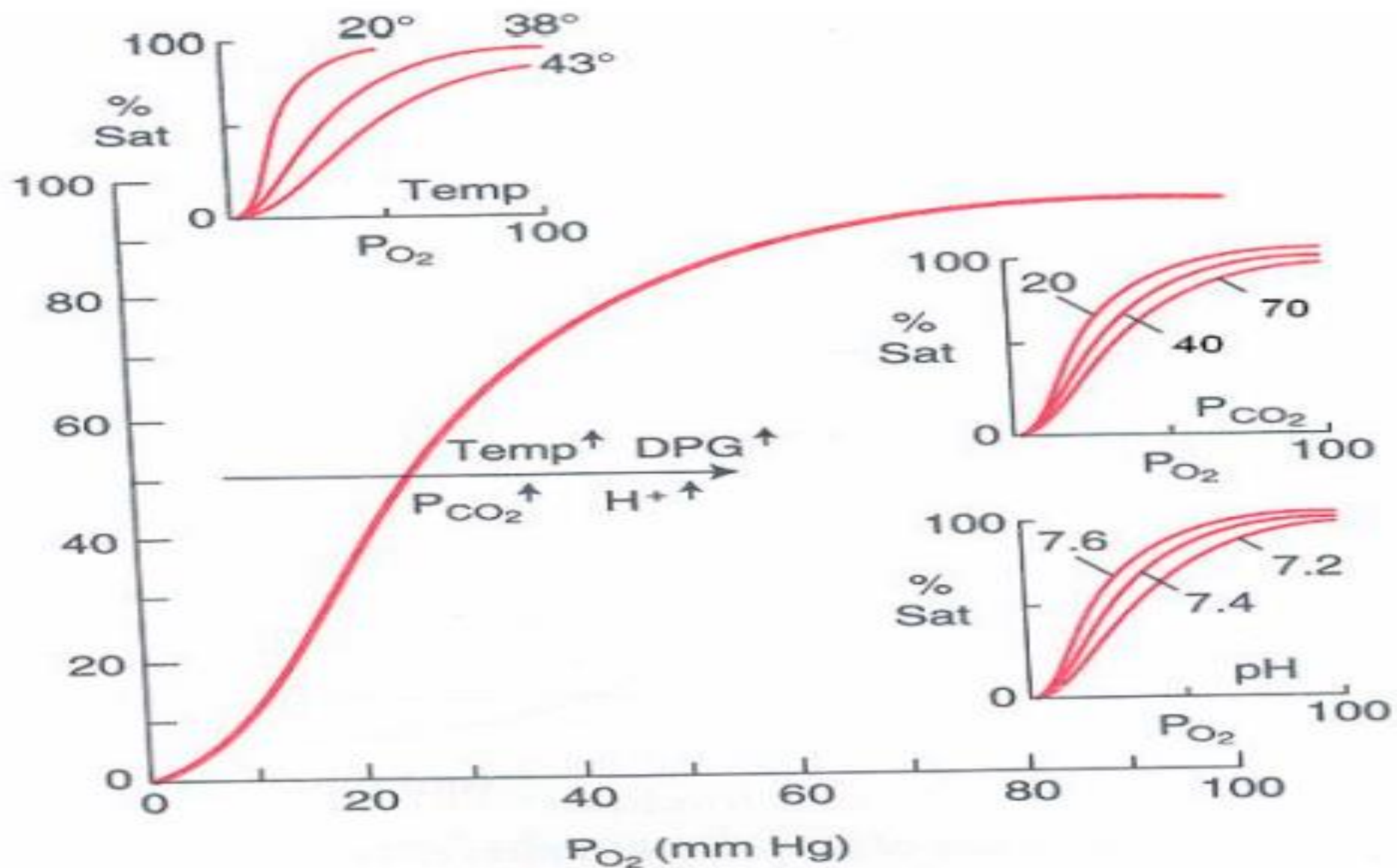
■ **Shift the Curve to the Right** because of

- ☐ A decrease in p^H
- ☐ An increase in carbon dioxide
- ☐ An increase in temperature
- ☐ substance 2,3-bisphosphoglycerate

☒ Results in; **decreased** the ability of hemoglobin to **binds with oxygen** (hemoglobin **increases** the release of O_2)



% Hb saturation



**Effect of P_{O_2} , pH, P_{CO_2} , and Temperature
on
Oxygen-hemoglobin dissociation curve**

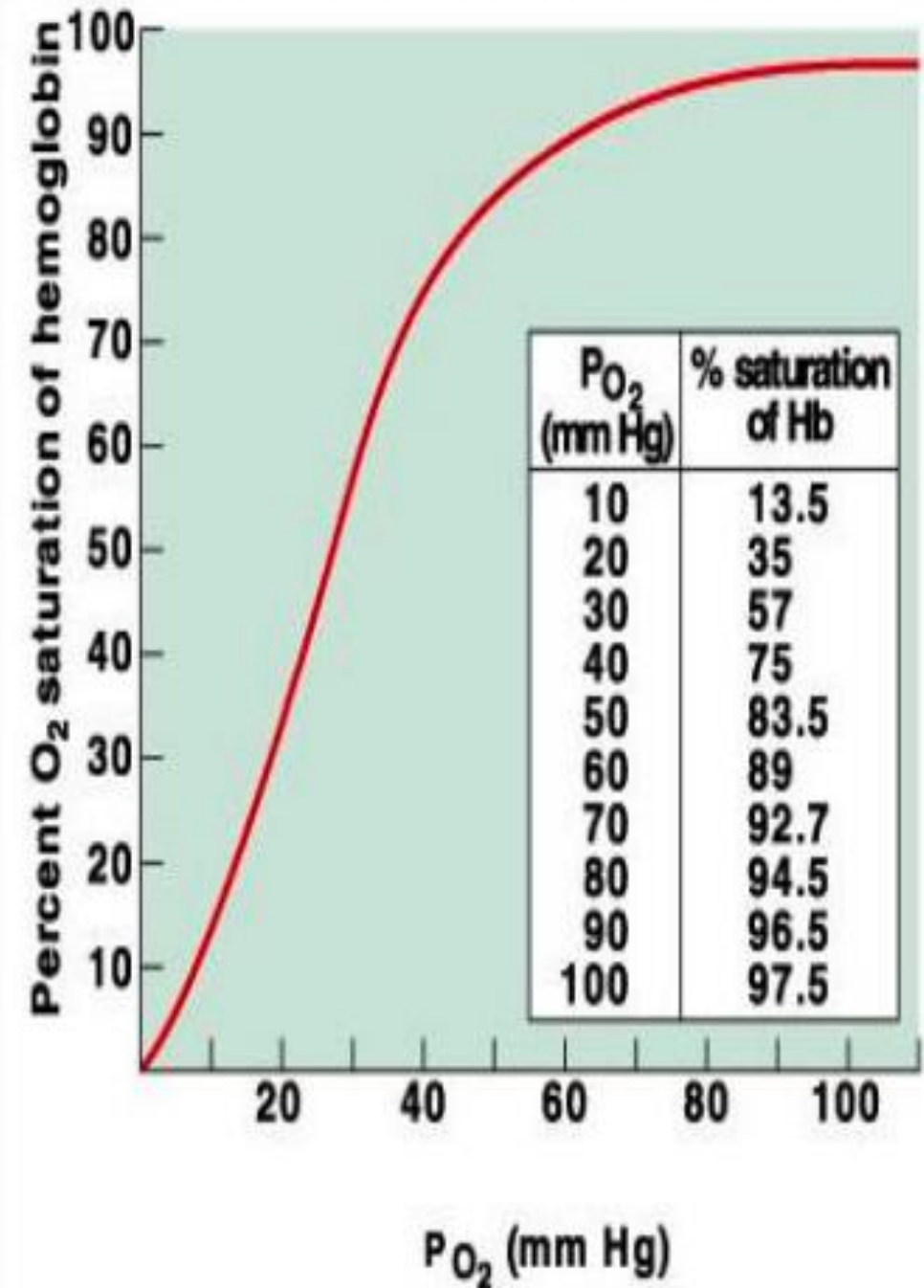
Effect of P_{O_2}

■ P_{O_2} in the blood leaving the **pulmonary capillaries** is normally **104 mm Hg**. At that partial pressure, **hemoglobin is 98% saturated**.

□ **Decreases in P_{O_2} in the pulmonary capillaries** have a relatively **small effect on hemoglobin saturation**

⇒ Even if the P_{O_2} **decreases from 104 mm Hg to 60 mm Hg**, hemoglobin is still 90% saturated.

□ **Hemoglobin** is very effective at picking up **oxygen** in the **lungs**, even if the **P_{O_2} in the pulmonary capillaries decreases significantly**.

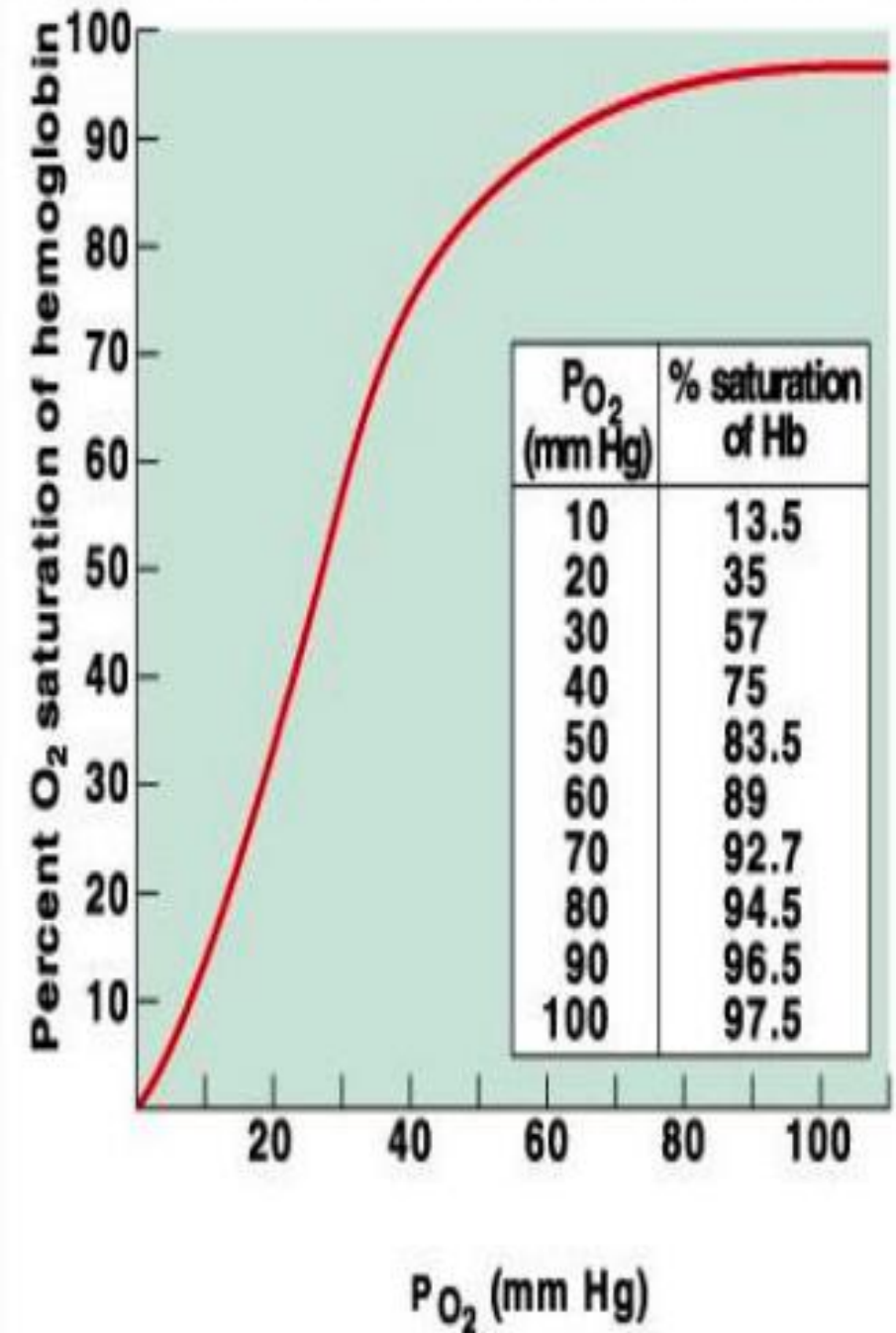


❑ **In the tissues**, relatively small change in **Po₂** results in **large change in hemoglobin saturation**.

⇒ For example, during vigorous exercise, the **Po₂** in skeletal muscle capillaries decline to **15 mm Hg**.

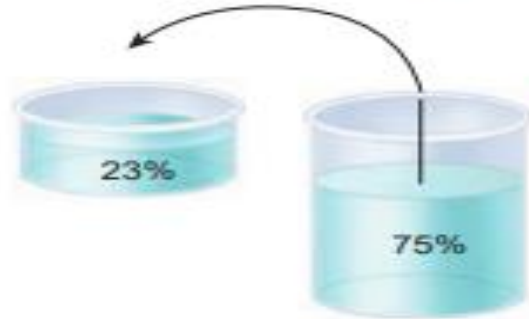
⇒ At a **Po₂ of 15 mm Hg, hemoglobin is only 25% saturated**, resulting in the **release of 73% (98% – 25%) of the oxygen** picked up in the **lungs**.

⇒ Thus, as tissues use more oxygen, hemoglobin releases more oxygen to those tissues.

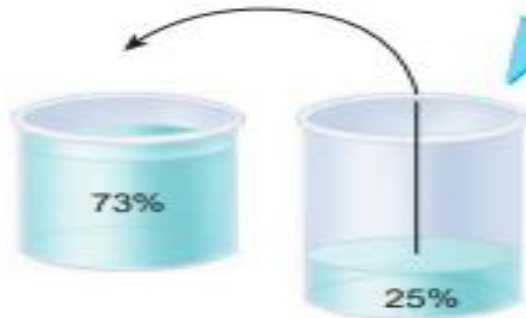


The ability of hemoglobin to pick up oxygen in the lungs and release it in the tissues is like a glass filling and emptying

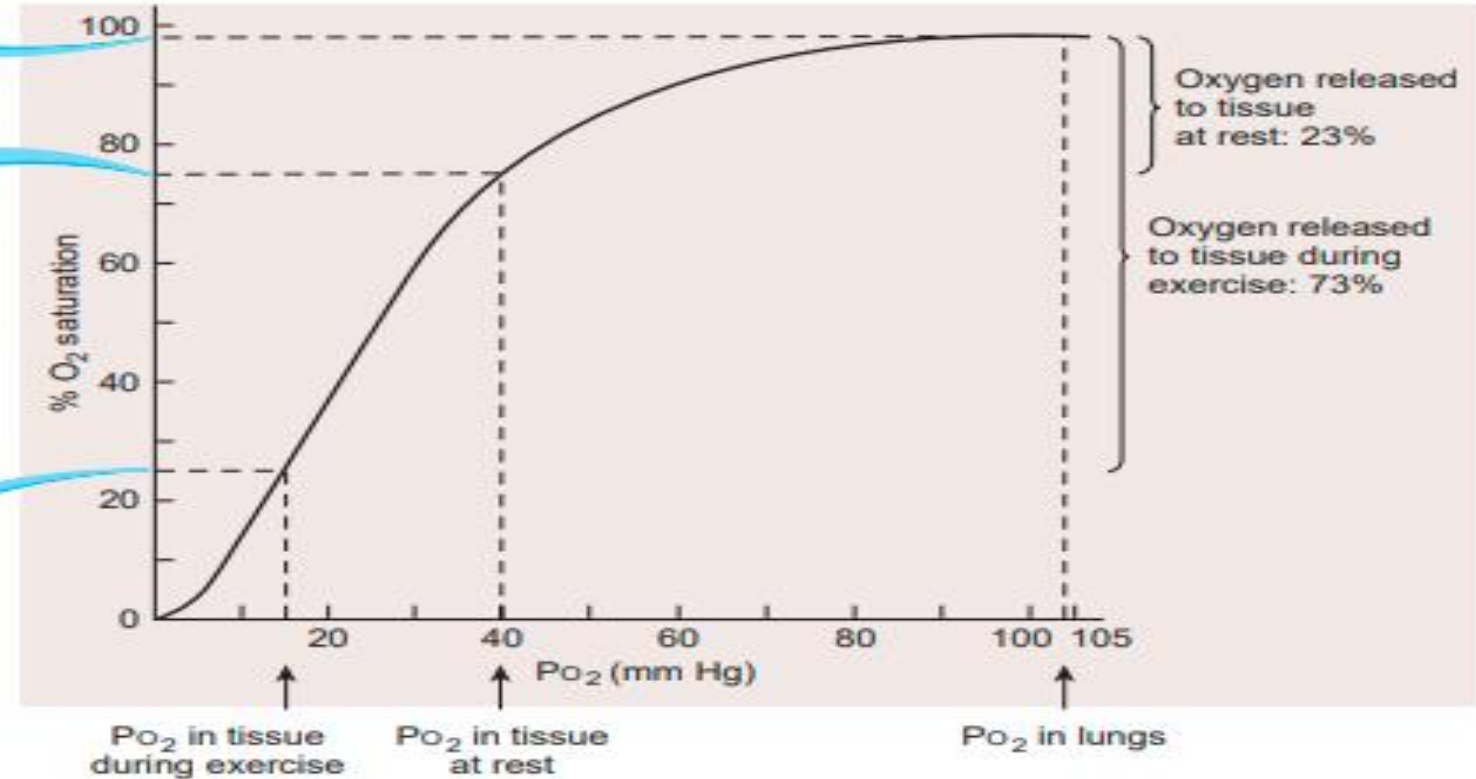
(a) **Lungs.** At a P_{O_2} of 104 mm Hg in the lungs, hemoglobin is 98% saturated with oxygen. It is like nearly filling a glass.



(b) **Tissue at rest.** At a P_{O_2} of 40 mm Hg in a tissue at rest, hemoglobin is 75% saturated with oxygen and 23% of the oxygen picked up in the lungs is released to the tissue. It is like partially emptying the glass.



(c) **Tissue during exercise.** At a P_{O_2} of 15 mm Hg in a tissue during exercise, hemoglobin can be 25% saturated with oxygen and 73% of the oxygen picked up in the lungs is released to the tissue. It is like emptying most of the glass.



Oxygen-Hemoglobin Dissociation Curve

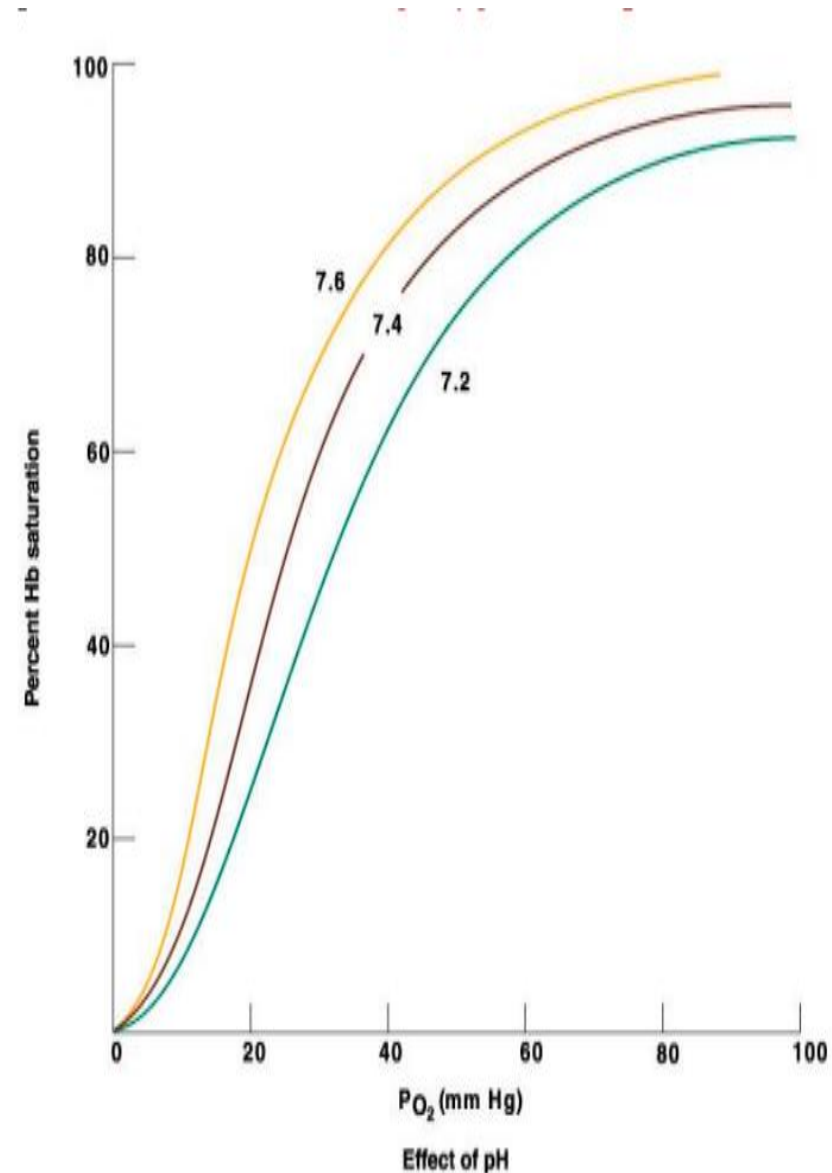
Effect of pH

As the **pH of blood declines**, the amount of oxygen bound to hemoglobin also declines.

- ❑ This occurs because **decreased pH results from an increase in H^+** , and the H^+ combine with hemoglobin molecule and **change its three-dimensional structure**
- ❑ Causing a **decrease in the hemoglobin's ability to bind oxygen**.

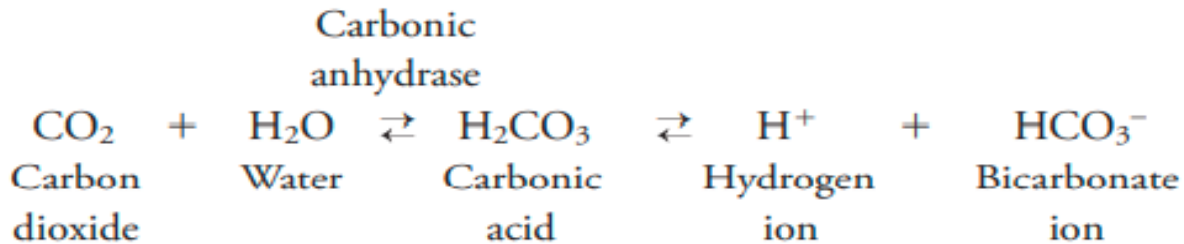
Conversely, an **increase in blood pH results in an increase in hemoglobin's ability to bind oxygen**.

- ❑ The effect of pH on the oxygen–hemoglobin dissociation curve is called the **Bohr effect**.



Effect of CO₂

■ An **increase in P_{CO2}** also decreases hemoglobin's ability to bind oxygen because of the **effect of carbon dioxide on pH**.



□ As **CO₂** levels **increase**, more **H⁺** are produced, and the **pH declines**. As **carbon dioxide** levels **decline**, the reaction proceeds in the opposite direction, resulting in a **decrease in H⁺ concentration** and an **increase in pH**.

⇒ Changes in carbon dioxide levels **in directly** produce a Bohr effect by altering pH

❑ In addition, carbon dioxide can **directly affect** hemoglobin's ability to bind oxygen, to a small extent.

■ As blood passes through **tissue capillaries**, carbon dioxide enters the blood from the **tissues**. As a consequence, **blood carbon dioxide levels increase, pH decreases**, and **hemoglobin has less affinity for oxygen in the tissue capillaries**.

■ When blood is returned to the **lungs** and passes through the **pulmonary capillaries**, carbon dioxide leaves the capillaries and enters the **alveoli**. As a result, **carbon dioxide levels in the pulmonary capillaries are reduced, pH increases**, and **hemoglobin's affinity for oxygen increases**.

Effect of Temperature

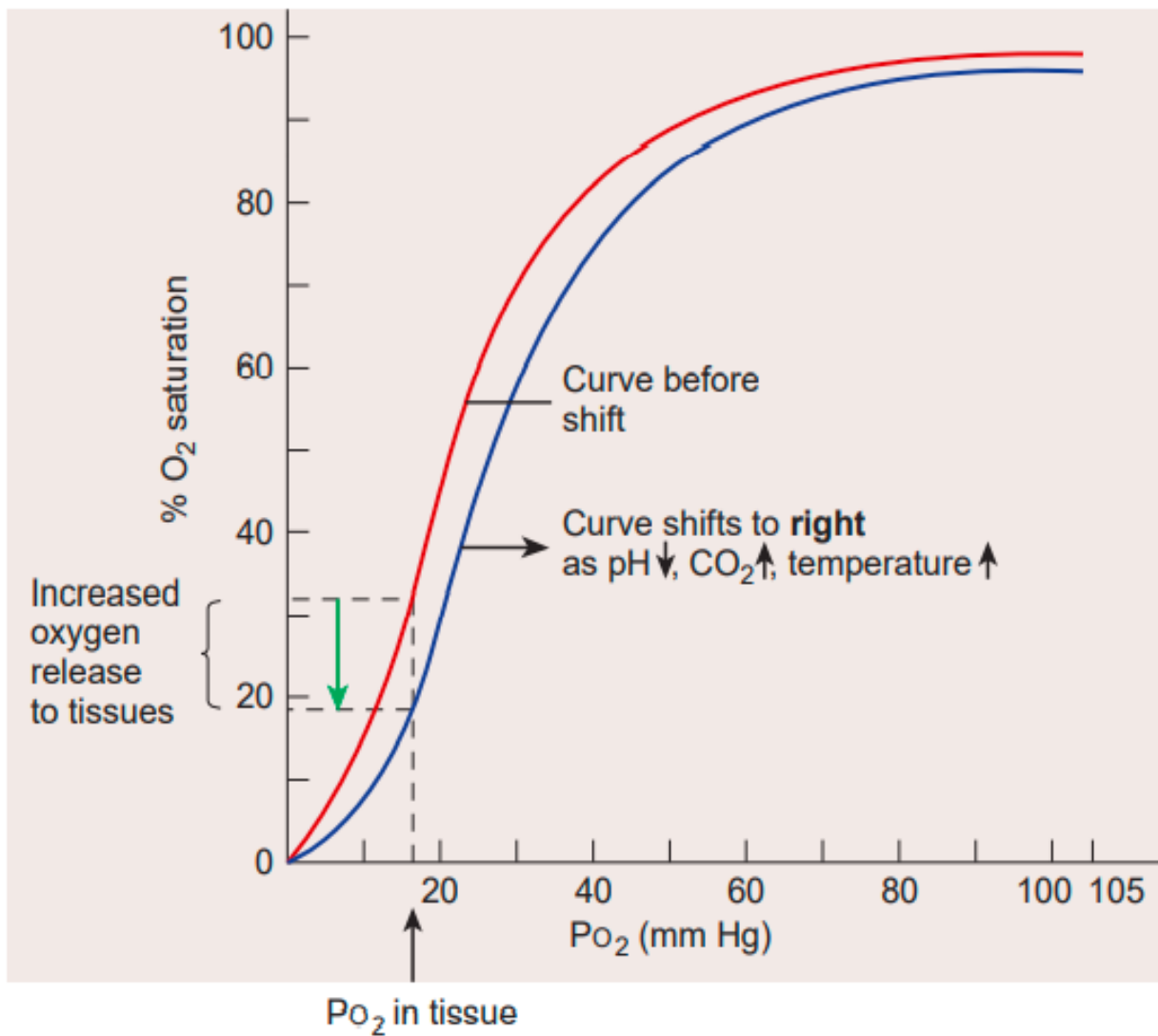
■ An **increase in temperature also decreases oxygen's tendency to remain bound to hemoglobin.**

□ Elevated temperatures resulting from **increased metabolism**, therefore, increase the amount of oxygen released into the tissues by hemoglobin.

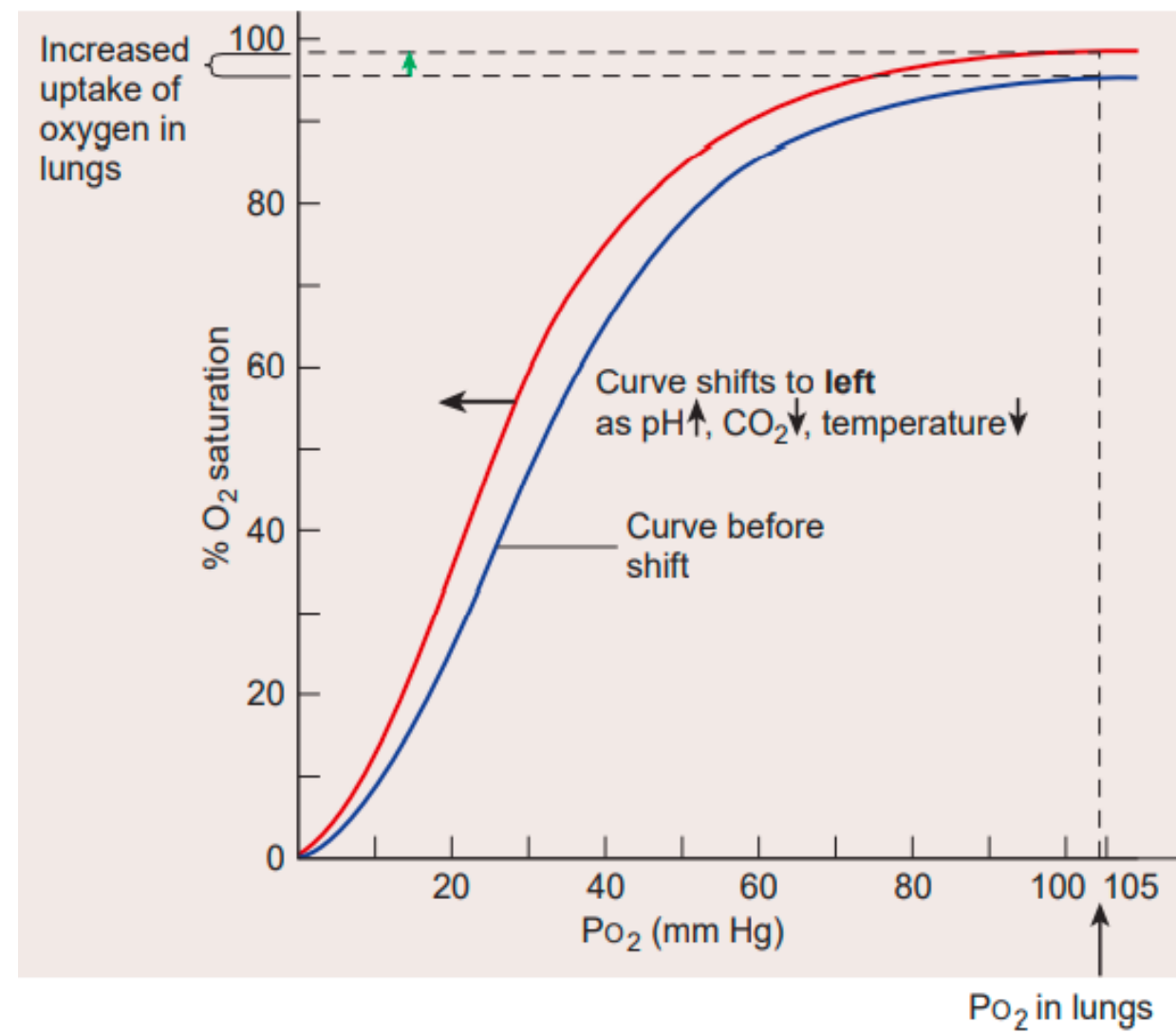
❑ In the tissues, the hemoglobin's affinity for oxygen decreases, the **oxygen– hemoglobin dissociation curve is shifted to the right**, and hemoglobin releases more oxygen.

❑ In the Lungs, however, the **curve shifts to the left** because of the **lower carbon dioxide levels, lower temperature, and lower lactic acid levels**.

☒ Hemoglobin's affinity for oxygen, therefore, increases, and it becomes easily saturated.



- ❑ **In the tissues**, the oxygen–hemoglobin dissociation curve **shifts to the right**. As pH decreases, PCO₂ increases, or temperature increases, the curve (red) shifts to the right (blue), resulting in an **increased release of oxygen**.



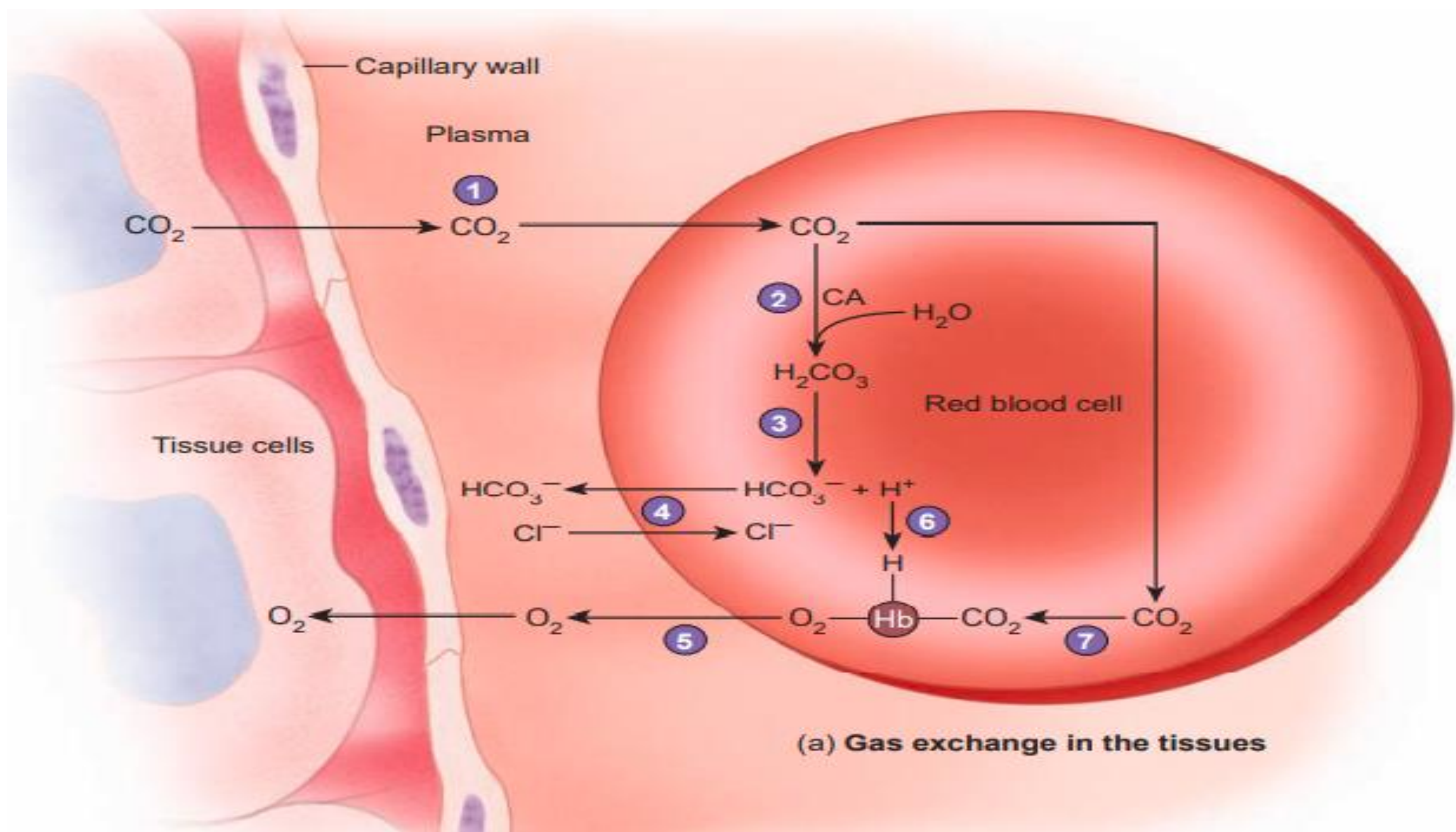
- ❑ **In the lungs**, the oxygen–hemoglobin dissociation curve **shifts to the left**. As pH increases, PCO₂ decreases, or temperature decreases, the curve (blue) shifts to the left (red), resulting in an **increased ability of hemoglobin to pick up oxygen**.

Transport of Carbon Dioxide

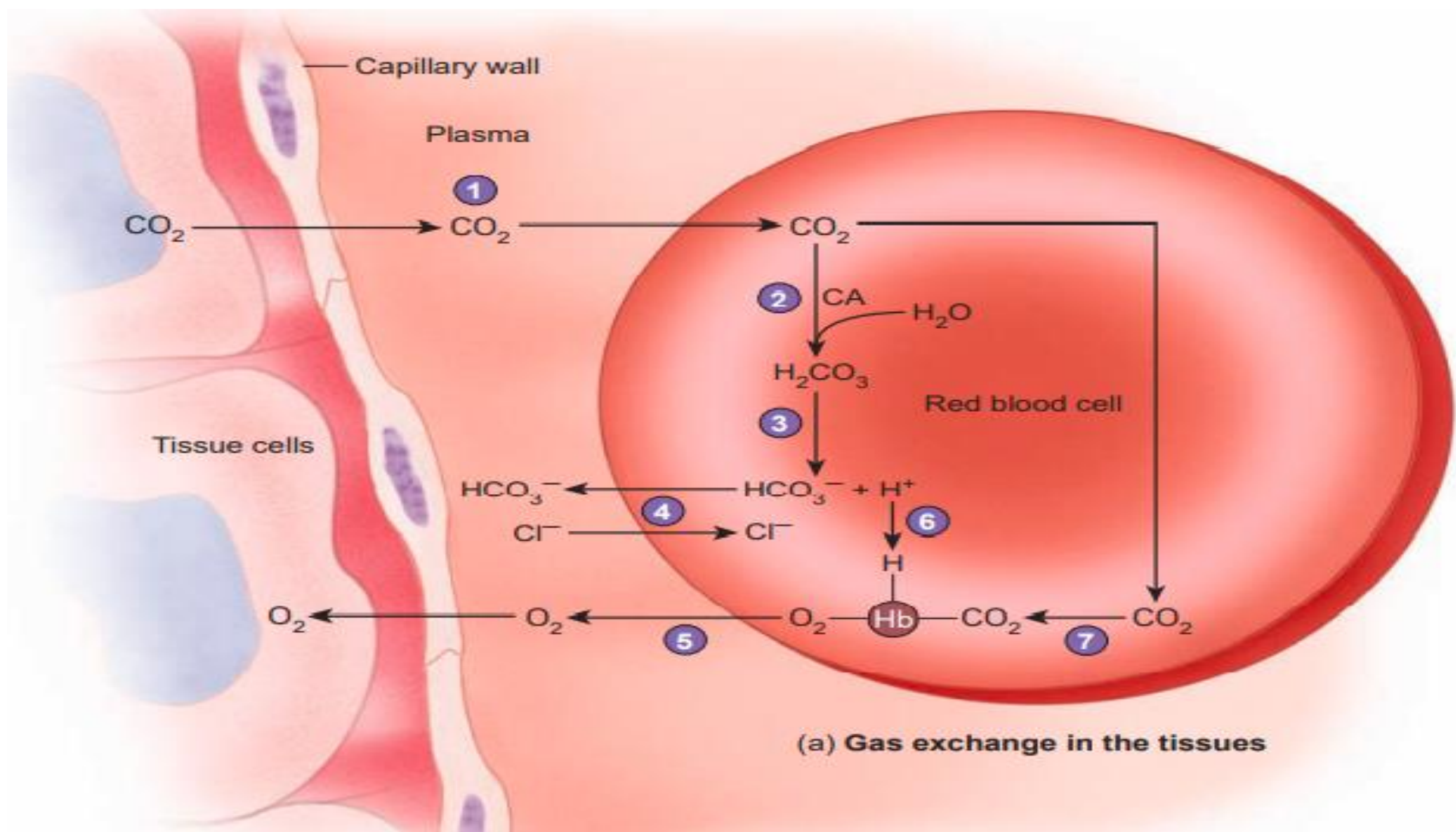
- ❑ Carbon dioxide is transported in the blood in three ways:
 1. By dissolved in the plasma - 7%
 2. In the form of HCO_3^- – 70%
 3. Bound to hemoglobin (Carbaminohemoglobin) - 23%

■ Carbon Dioxide Exchange in Tissues

- ❑ Carbon dioxide diffuses from **the tissues into plasma of blood**. Most of the carbon dioxide diffuses into **red blood cells**, but approximately 7% of it is transported by dissolved in the **plasma**.
- ❑ **Carbon dioxide inside red blood cells** reacts with water to form **carbonic acid**, a reaction catalyzed by carbonic anhydrase. **Carbonic acid** then dissociates to form HCO_3^- and H^+ . Approximately **70% of carbon dioxide is transported in the form of HCO_3^-** .



- ❑ **Removing HCO_3^-** – from inside the red blood cells **promotes carbon dioxide transport** because, as HCO_3^- – concentration decreases, more carbon dioxide combines with water to form **additional HCO_3^-** – and **H^+** .
- ❑ In a process called **chloride shift** , antiporters exchange Cl^- for HCO_3^- .
 - ⇒ This exchange maintains electrical balance in the red blood cells and plasma as HCO_3^- – diffuse out of, and Cl^- diffuse into, **red blood cells**.
- ❑ **Hydrogen ions bind to hemoglobin** . As a result, **three effects** are produced:
 1. **Transport of carbon dioxide increases** because, as H^+ concentration decreases, more carbon dioxide combines with water to form **additional HCO_3^-** – and **H^+**
 2. **pH inside the red blood cells does not decrease** because hemoglobin is a buffer, **preventing an increase in H^+ concentration**
 3. **Affinity of hemoglobin for oxygen decreases**.



(a) Gas exchange in the tissues

⇒ **Hemoglobin's decreased affinity for oxygen** results in; a shift of the oxygen–hemoglobin curve to the right and **increase the release of oxygen from hemoglobin** in tissue capillaries.

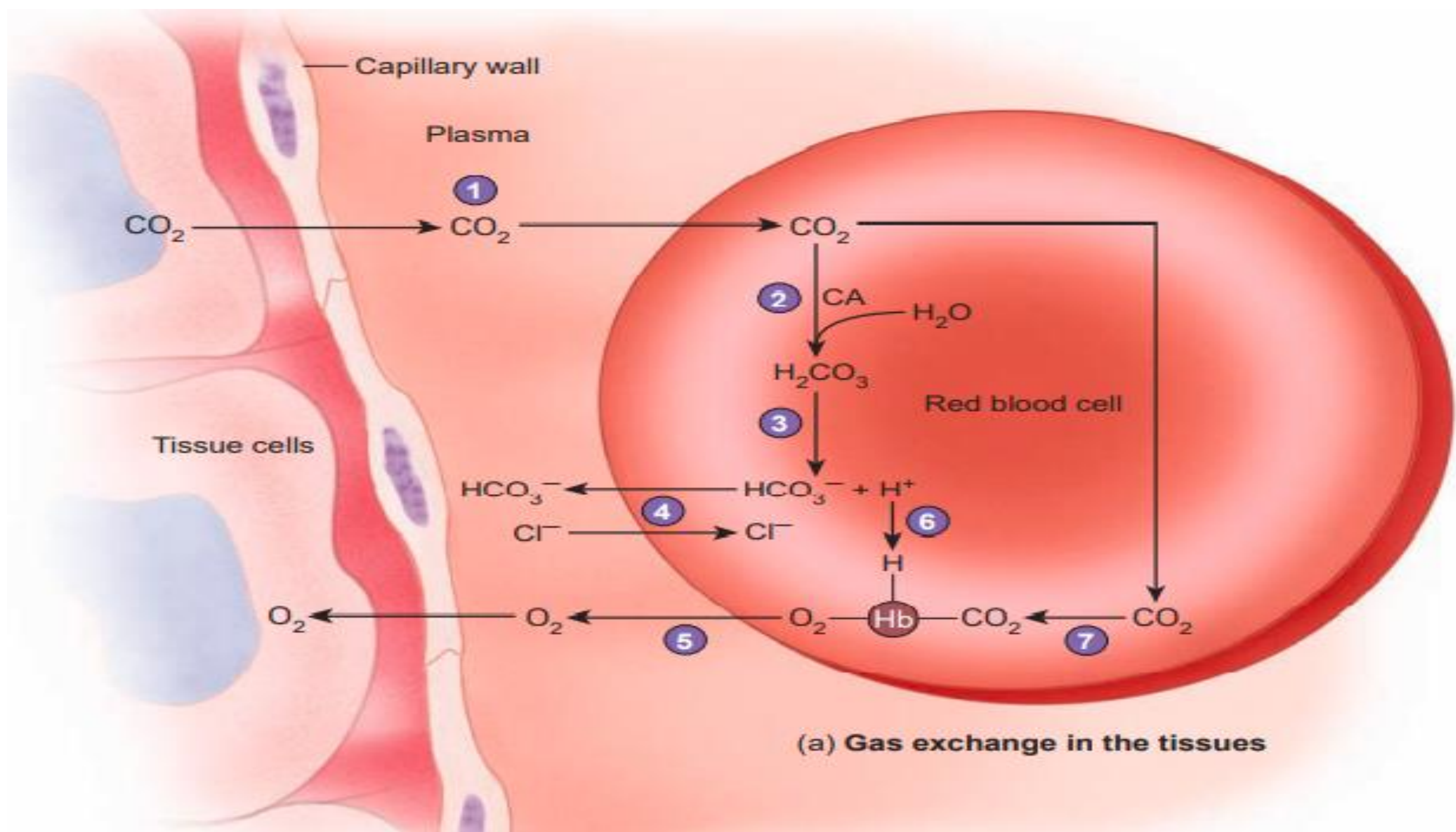
□ Approximately **23%** of the carbon dioxide is transported bound to **hemoglobin**.

⊗ The ability of **carbon dioxide bind to hemoglobin** is affected by the **amount of oxygen bound to hemoglobin**.

⇒ The **smaller amount of oxygen** bound to hemoglobin, the **greater amount of carbon dioxide** can bind to it, and vice versa. This relationship is called **Haldane Effect**.

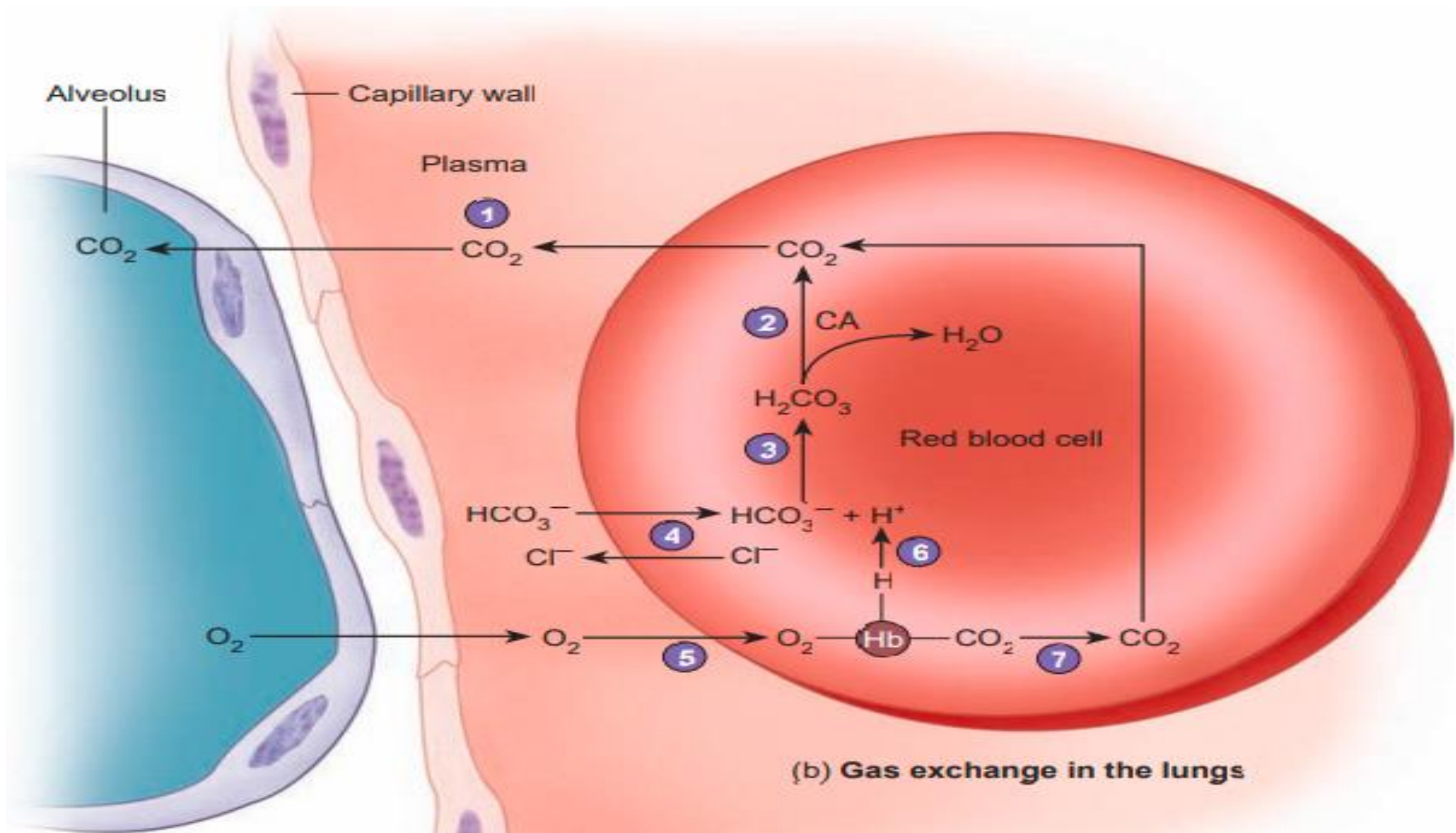
■ **In Tissues**, as hemoglobin releases oxygen, the **hemoglobin has an increased ability to pick up carbon dioxide**

1. **In the tissues**, carbon dioxide (CO_2) diffuses into the **plasma and into red blood cells**. Some of **the carbon dioxide remains in the plasma**.
2. **In red blood cells**, carbon dioxide reacts with water form carbonic acid (H_2CO_3) in a reaction catalyzed by the enzyme carbonic anhydrase (CA).
3. Carbonic acid dissociates to form **bicarbonate ions (HCO_3^-)** and **hydrogen ions (H^+)**.
4. In the **chloride shift**, as **HCO_3^- move out of the red blood cells**, electrical neutrality is maintained by the **movement of chloride ions (Cl^-) into them**.
5. **Oxygen (O_2) is released from hemoglobin (Hb)**. Oxygen diffuses out of **red blood cells and plasma into the tissue**.
6. Hydrogen ions combine with hemoglobin, which promotes the **release of oxygen from hemoglobin (Bohr effect)**.
7. **Carbon dioxide combines with hemoglobin**. Hemoglobin that **has released oxygen** readily combines with carbon dioxide (**Haldane effect**).



■ Carbon Dioxide Exchange in the Lungs

- ❑ **Carbon dioxide** diffuses from **red blood cells** and **plasma** into the **alveoli** . As carbon dioxide levels in the red blood cells decrease, carbonic acid is converted to carbon dioxide and water.
- ☒ **In response**, HCO_3^- join with H^+ to form carbonic acid. As **HCO_3^- and H^+ concentrations decrease because of this reaction, HCO_3^- enter red blood cells in exchange for Cl^- and H^+ are released from hemoglobin.**
- ☒ **Hemoglobin picks up oxygen in pulmonary capillaries** b/c of increased Po_2 .
- ☒ The release of H^+ from hemoglobin increases hemoglobin's affinity for oxygen
- ☒ **Oxygen** from the **alveoli diffuses into the pulmonary capillaries** and **into the red blood cells**, and it binds with hemoglobin.
- ☒ **Carbon dioxide** is released from hemoglobin and **diffuses out of the red blood cells into the alveoli**. As hemoglobin binds to oxygen, it more readily releases carbon dioxide (Haldane effect)



(b) Gas exchange in the lungs

1. **In the lungs**, carbon dioxide (CO_2) diffuses from **red blood cells and plasma into the alveoli**.
2. Carbonic anhydrase catalyzes the formation of CO_2 and H_2O from **H_2CO_3** .
3. **Bicarbonate ions** and **H^+ combine** to replace H_2CO_3 .
4. **In the chloride shift**, as HCO_3^- move into red blood cells, electrical neutrality is maintained by the **movement of chloride ions (Cl^-) out of them**.
5. **Oxygen** diffuses into the **plasma and into red blood cells**. Some of the oxygen remains in the plasma. **Oxygen binds to hemoglobin**.
6. **Hydrogen ions are released from hemoglobin**, which promotes the uptake of oxygen by hemoglobin (**Bohr effect**).
7. **Carbon dioxide is released from hemoglobin**. Hemoglobin that is bound to oxygen readily **releases carbon dioxide** (**Haldane effect**).

Regulation of Respiration / Ventilation

Regulatory Mechanisms of Breathing

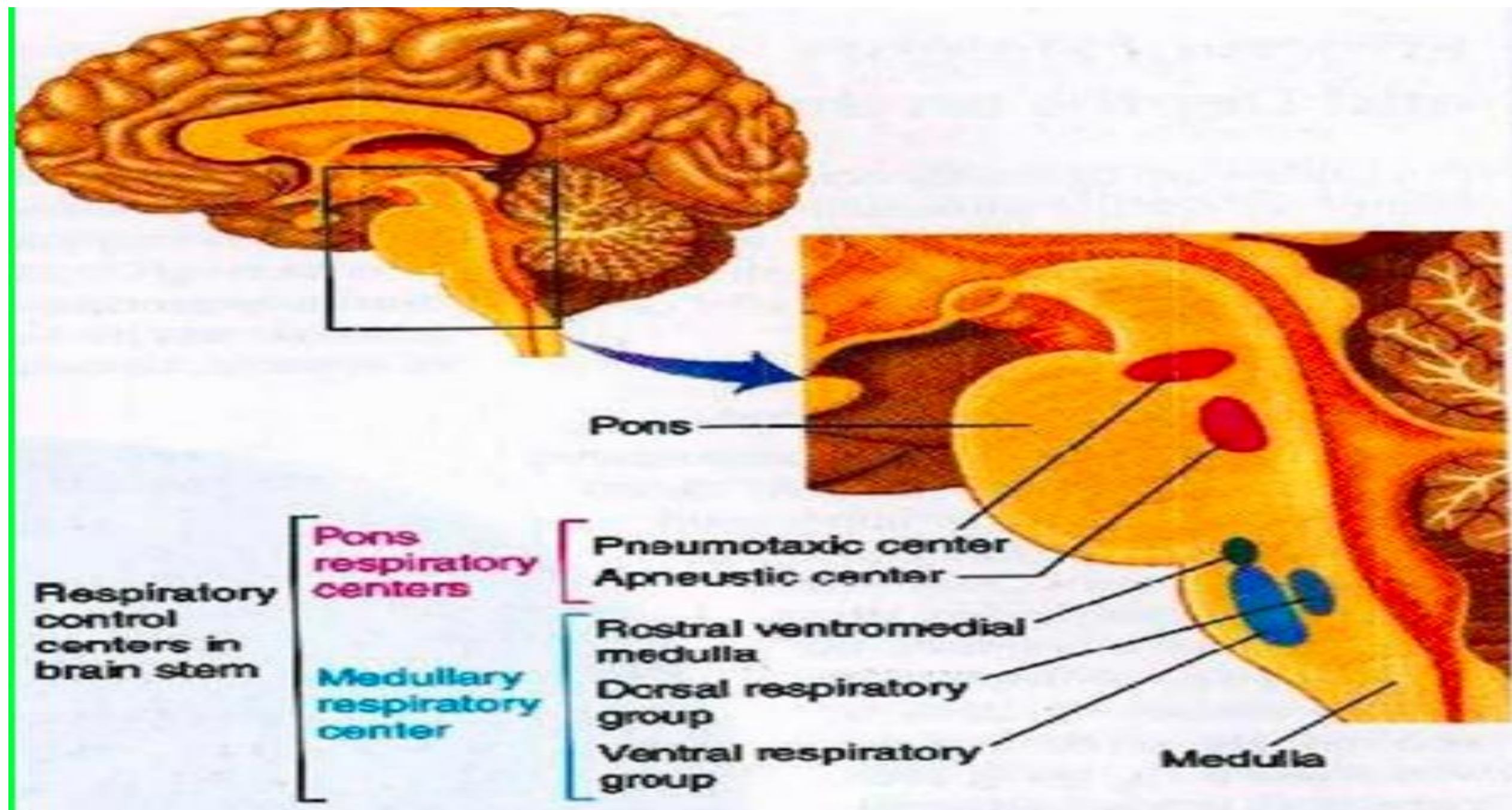
- ❑ Breathing regulatory mechanism has three basic components:
 1. Sensors receptors (Peripheral and Central chemoreceptors)
 2. Control centers (Brain stem and Cerebral Cortex)
 3. Effectors (Lungs, respiratory muscles)
- **Neural control** is both voluntary and involuntary
 - ❑ **Pons** and **Medulla** control the periodic nature of **involuntary respiration** (inspiration and expiration)
 - ⇒ Constitute the **respiratory centers**
 - ❑ **Cerebral Cortex** is responsible for coordinate **breathing with voluntary activities** and can override the function of brain stem **within physiological limits.**

Respiratory Centers

- ❑ **Brain Stem** forms **TWO** respiratory centers such as **medullary** and **pontine** respiratory centers

1. Medullary Respiratory Center

- ❑ Initiate and maintain the **spontaneous breathing patterns** (**Rhythmic nature of respiration originates**)
- ❑ Consist of **Ventral Respiratory Group** and **Dorsal Respiratory Group**
 - ❑ VRG and DRG constitute the **primary respiratory centers**
- ❑ **VRG** consists of a long column of **EXPIRATORY NEURONES**
- ❑ **DRG** consists of **INSPIRATORY NEURONS**

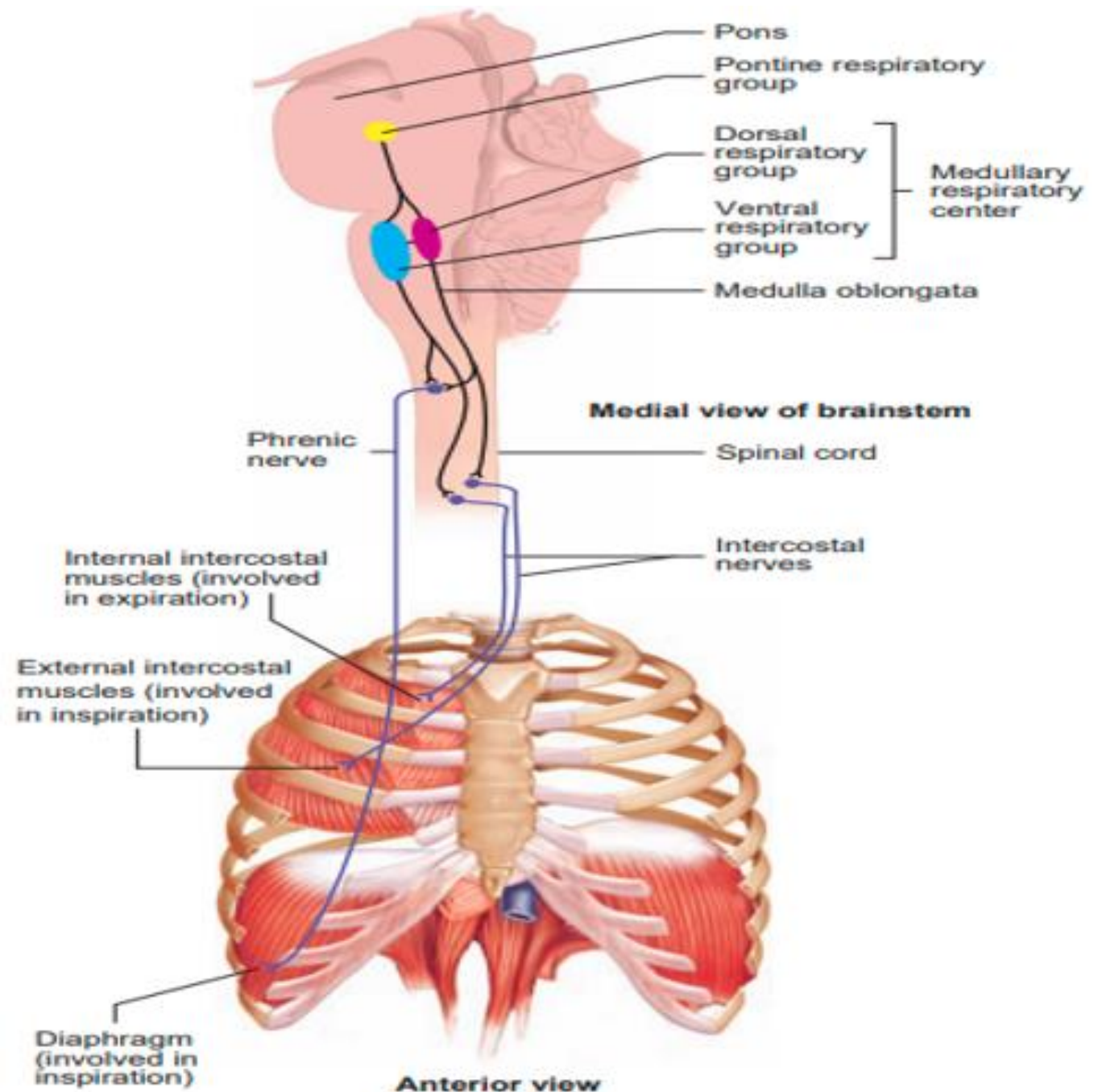


❑ **Dorsal Respiratory Group** is a major respiratory rhythm generator

⇒ **DRG** receives sensory input from peripheral chemoreceptors and stretch receptors in the lungs. It communicates with the ventral respiratory group and primarily promotes contraction of **muscles during inspiration** e.g., stimulates the diaphragm

❑ **Ventral Respiratory Group (VRG)**

⇒ promotes muscle contractions **during expiration**. e.g., stimulates the internal intercostal and abdominal muscles.



2. Pontine Respiratory Center

- ⇒ Secondary respiratory center
- ⇒ Has connections with the medullary respiratory center and **play a major role in switching between inspiration and expiration.**
- ⇒ Influences output from medullary respiratory center

□ It consist of:

■ **Pneumotaxic Neuron** (Inhibitory Center)

- ⇒ “Switch off” I-neurons (limits duration of inspiration)

■ **Apneustic neuron** (gasping Center)

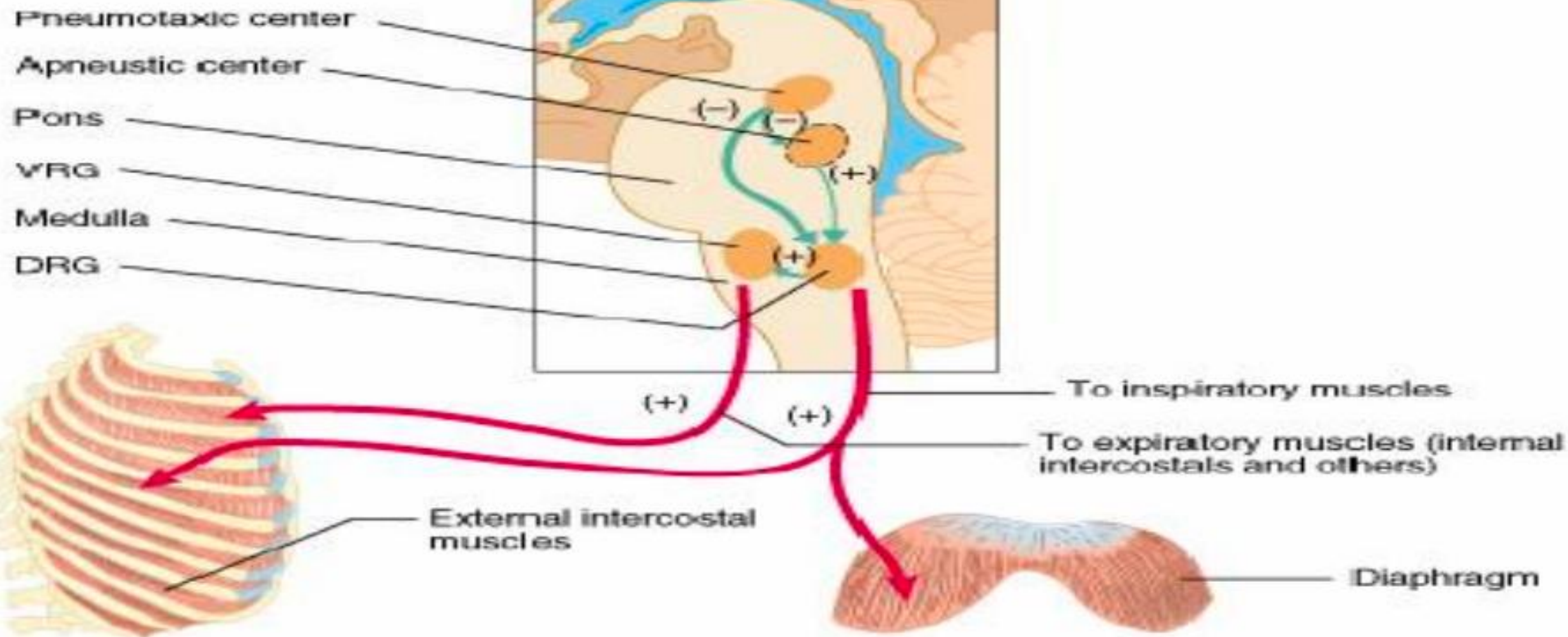
- ⇒ Prevent I-neurons from being switched off

□ Pneumotax is more dominant than apneustic ⇒ to allow smooth breathing



(+) = Positive effect
(stimulation)

(-) = Negative effect
(inhibition)



Chemoreceptors

❑ **Respiratory System** maintains blood **O₂ and Co₂ concentrations** and **blood pH** within a normal range. Deviation from their normal range has a marked **influence on respiratory movements**.

⇒ The effect of changes in **O₂** and **Co₂ concentrations** and **pH** is superimposed on **Neural Mechanisms** that establish rhythmic ventilation

■ **Chemoreceptors** regulate respiration in response to; **changes in H⁺ concentration**, **changes in Po₂**, and **PCo₂**

1. Peripheral Chemoreceptors
2. Central Chemoreceptors

Influence of Chemical Factors on Respiration

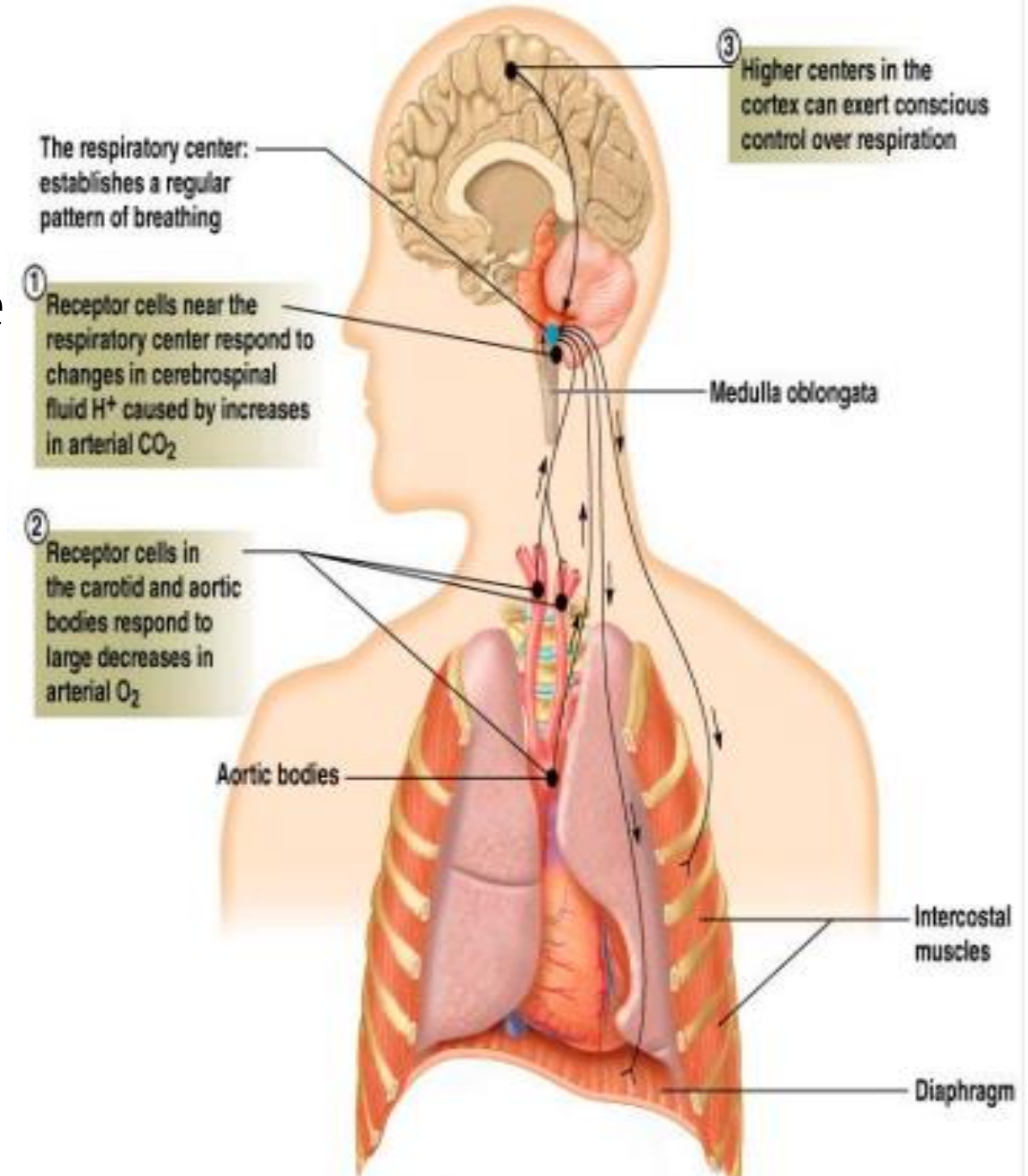
Chemical Factor	Effect on the Peripheral Chemoreceptors	Effect on the Central Chemoreceptors
$\downarrow P_{O_2}$ in the Arterial Blood	Stimulates only when the arterial P_{O_2} has fallen to the point of being life threatening (<60 mm Hg); is an emergency mechanism	Directly depresses the central chemoreceptors and the respiratory center itself when < 60 mm Hg
$\uparrow P_{CO_2}$ in the Arterial Blood ($\uparrow H^+$ in the Brain ECF)	Weakly stimulates	Strongly stimulates; is the dominant control of ventilation (Levels > 70 – 80 mm Hg directly depress the respiratory center and central chemoreceptors)
$\uparrow H^+$ in the Arterial Blood	Stimulates; is important in acid–base balance	Does not affect; cannot penetrate the blood–brain barrier

■ **Peripheral Chemoreceptors** are found in the carotid and aortic bodies.

- ❑ Peripheral Chemoreceptors are more sensitive to when the **$P_{O_2} \downarrow \downarrow$** and **$H^+$ concentration in artery $\uparrow$**

■ **Central Chemoreceptors** are located in the chemosensitive area of medulla oblongata.

- ❑ Central Chemoreceptors are more sensitive when the **p^H in cerebrospinal fluid $\downarrow$** and **$PCO_2 \uparrow$**



Effect of pH

❑ **Central Chemoreceptors** and **Peripheral Chemoreceptors** respond to changes in **blood pH**.

☒ Central chemoreceptors respond **indirectly** to changes in blood pH, whereas the peripheral chemoreceptors respond **directly**.

⇒ **Hydrogen ions** do not cross the **blood–brain barrier** to affect the central chemoreceptors, but they easily **cross from the blood to the peripheral chemoreceptors**.

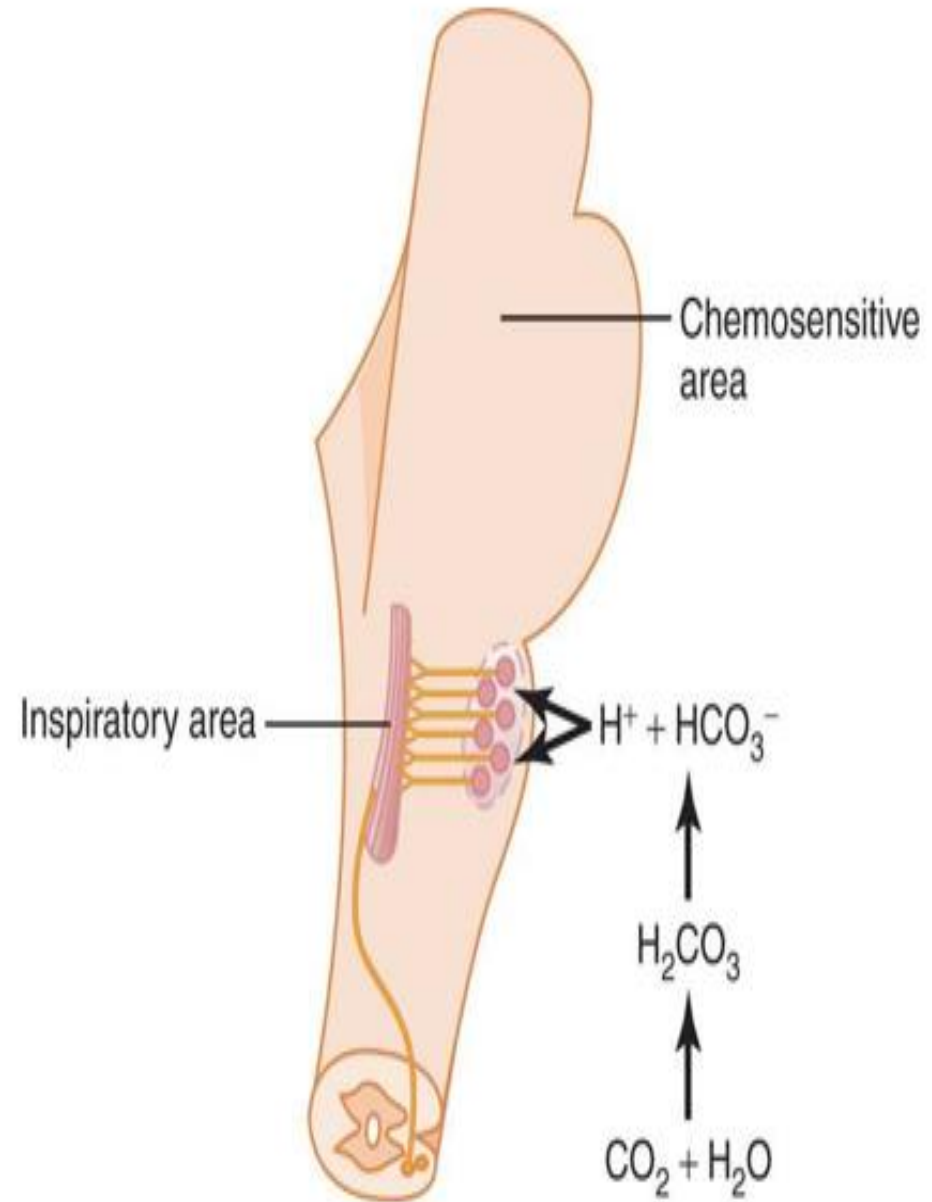
■ **Central Chemoreceptors** detects **changes in blood pH** through **changes in blood carbon dioxide**, which easily diffuses across the blood–brain barrier.

❑ For example, if blood **carbon dioxide levels increase**, carbon dioxide diffuses across the blood–brain barrier into the cerebrospinal fluid.

⇒ Carbon dioxide combines with water to form carbonic acid, which dissociates into H^+ and HCO_3^- .

⇒ The **increased concentration of H^+** lowers the pH and stimulates the central chemoreceptors, which then stimulates the respiratory center, **resulting in** a greater rate and depth of breathing.

❑ **Consequently**, carbon dioxide levels decrease as carbon dioxide is eliminated from the body and **blood pH increases** to normal levels.



❑ **Changes in carbon dioxide levels** can **changes the pH**, the respiratory system plays an important role in **acid–base balance**

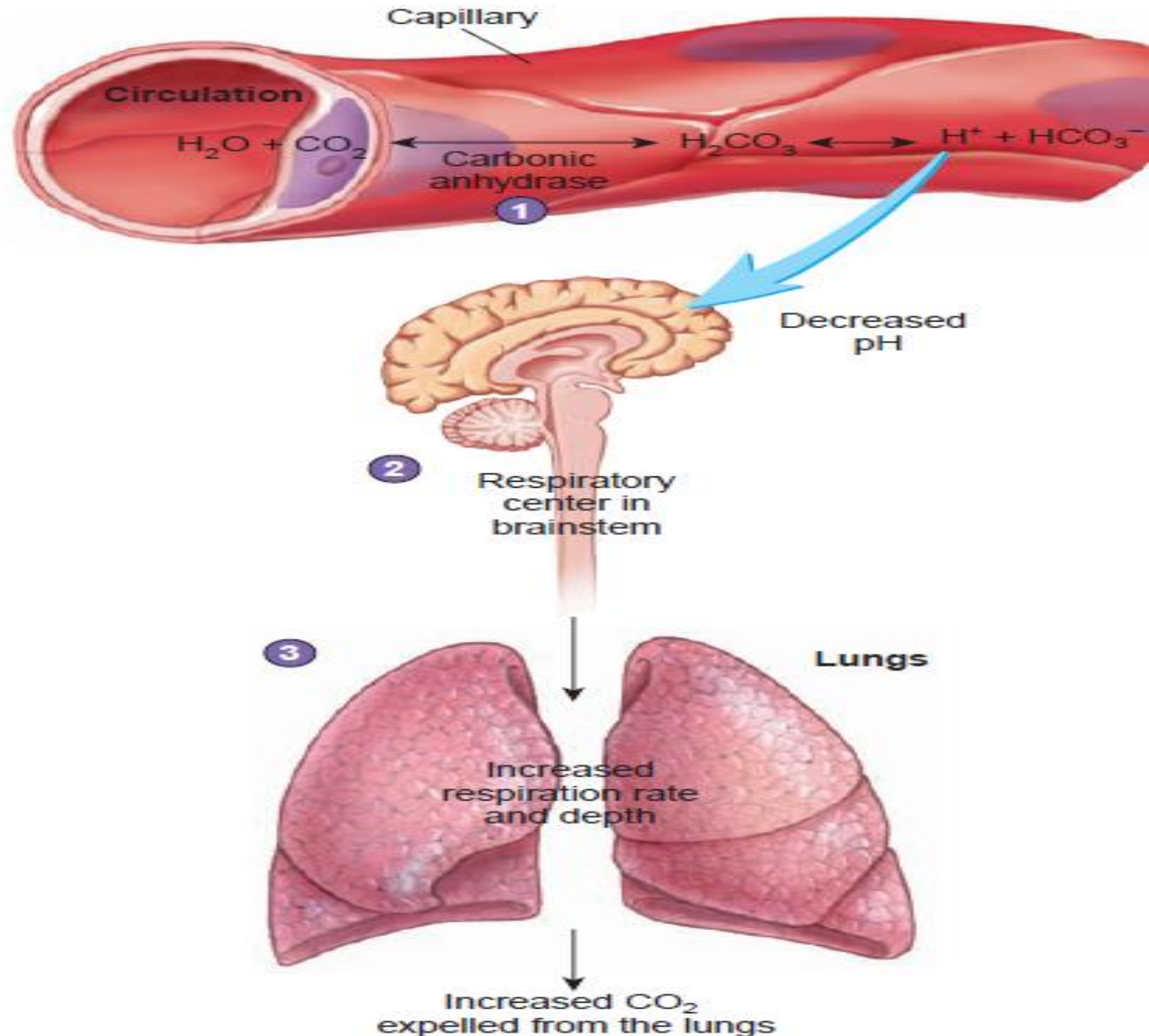
☒ As **carbon dioxide concentration increases**, carbon dioxide combines with water to form H_2CO_3 . Carbonic acid then dissociate to form **H^+** and HCO_3^- , which **decreases blood pH**



■ **Decreased in blood pH, STIMULATE** neurons in the respiratory center and cause the **rate and depth of ventilation to increase.**

⇒ Increased rate and depth of ventilation eliminated the CO_2 from the body through the lungs

⇒ As carbon dioxide levels decline, the concentration of H^+ decreases, which **increases blood pH**



- ✗ If **carbon dioxide levels decline**, however, the reaction shifts in the opposite direction so that H^+ and HCO_3^- decrease, which **increases blood pH**



■ **Increased in blood pH, INHIBIT** neurons in the respiratory center and cause the rate and depth of ventilation to decrease.

- ⇒ Decreased rate and depth of ventilation cause less carbon dioxide to be eliminated from the body. The concentration of carbon dioxide in the blood increases
- ⇒ As carbon dioxide levels increase, the concentration of H^+ increases, which **decreases blood pH**

Effect of Carbon Dioxide

- ❑ Carbon dioxide levels in blood is a **major regulator of respiration** during resting condition & exercise condition
 - ☒ Even a small increase in carbon dioxide triggers a large increase in the rate and depth of ventilation. e.g., An $\uparrow$ in P_{CO_2} of 5 mm Hg causes an increase in ventilation of by 100%
- ❑ **During resting condition**
 - ☒ Central chemoreceptors are more important for the regulation of P_{CO_2} and pH than peripheral chemoreceptors
- **During intense exercise**, however, the peripheral chemoreceptors respond more rapidly to **changes in blood pH** than central chemoreceptors

Hering-Breuer Reflex

■ **Hering-Breuer reflex** limits the degree of inspiration proceeds and prevents over inflation of the lungs .

⇒ As the **lungs fill with air**, stretch receptors in the lungs stimulated. Action potentials from the lung stretch receptors propagate to the **medulla oblongata**, where they **inhibit the respiratory center and cause expiration**.

- ❑ **In infants**, the Hering-Breuer reflex plays an important role in regulating the basic rhythm of breathing and preventing over inflation of the lungs.
- ❑ **In adults**, however, the reflex is important **only when the tidal volume is large**, such as during heavy exercise



THE END